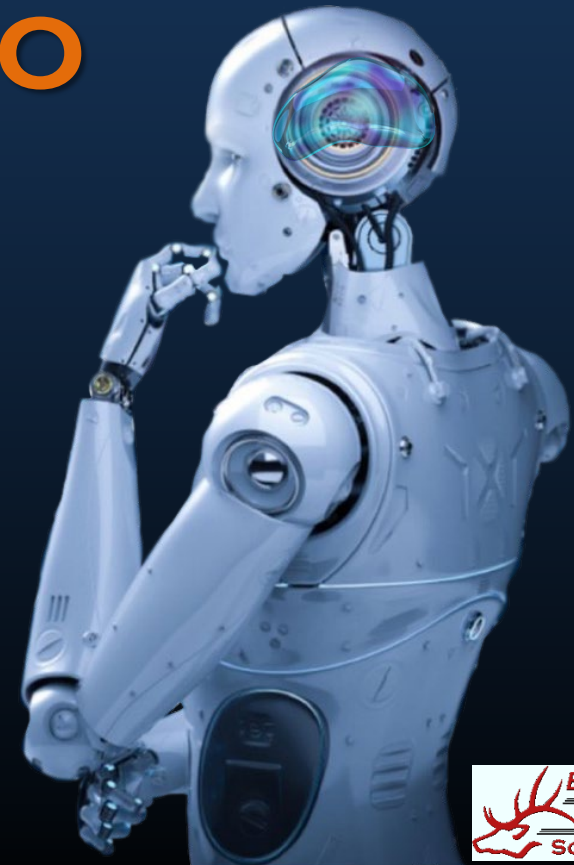


AI SKILLS FOR THE CISCO ACADEMY

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Instructor Professional Development
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WHY DOES AI MATTER?

Are Our Jobs in Jeopardy?

The landscape of networks is shifting, and it's time for us to **adapt**! Our approach to learning and teaching must **evolve** to truly resonate with today's learners.

Innovation = Job Security

CLI-only Era



This approach relies solely on manual command-line interface (CLI) operations for network management, where all configurations and troubleshooting are performed by human engineers, leading to potential inefficiencies and errors.



Automation



This approach utilizes scripts and tools to automate repetitive tasks in network management, enhancing efficiency and consistency while reducing the need for manual intervention, but it remains limited to predefined rules and lacks adaptive intelligence.



AI-Augmented Networking



This advanced approach integrates artificial intelligence and machine learning into network operations, enabling intelligent decision-making, predictive analytics, and autonomous management, which significantly enhances efficiency, security, and the ability to proactively address network issues.

What is AI in Networking?



Traditional Networking Tasks:

- Manually monitor device logs
- Write and apply CLI configs
- Respond to issues after they occur
- Interpret error messages through trial and error
- Manually tune performance and adjust routes

AI-Augmented Networking:

- Detect and predict failures using real-time data
- Generate config suggestions or full command sets
- Identify anomalies before users report issues
- Explain technical errors in plain language
- Dynamically optimize traffic and bandwidth

AI doesn't replace you — it amplifies your expertise.

Understanding AI's Limits in Networking

- **Accuracy Depends on Data Quality**
 - AI relies heavily on accurate and complete data. Poor input data leads to unreliable outputs.
- **AI Doesn't Replace Expertise**
 - It amplifies but doesn't replace human judgment, especially for complex, nuanced decisions.
- **Potential for Incorrect or Misleading Outputs**
 - Generative AI may occasionally produce plausible but incorrect suggestions or configurations.
- **Requires Human Oversight and Verification**
 - Always verify AI-generated solutions before applying them in production networks.

AI and ML in the CCNA 1.1



6.4 Explain AI (generative and predictive) and machine learning in network operations

Generative AI

- Automated network configurations
- Security enhancement
- Data analysis for event prediction
- Network management automation

Predictive AI

- Network traffic prediction and analysis
- Proactive security detection and resolution
- Performance optimization
- Equipment failure prediction

Automation and Programmability make up 10% of the CCNA exam.

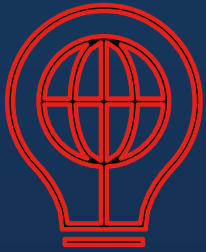
AI in Action

- **Generative AI:** Specializes in creating new content, such as visualizations, reports, or scripts, based on input data. It is beneficial for generating training materials, configuration templates, or summaries.
- **Predictive AI:** Focuses on analyzing historical data to predict future outcomes or trends. It is primarily used for forecasting and identifying potential risks or issues before they occur.
- **AI Agent:** Combines generative and predictive AI to autonomously pursue goals and complete tasks on behalf of users or other systems.
- **Agentic AI:** Goes beyond a single AI agent to autonomously reason, plan, and execute tasks.

AI in Action: How They Contribute to the Respective Tasks

Category	Generative AI	Predictive AI	Agentic AI
Monitoring	Generates visualizations or reports to summarize monitoring data.	Analyzes historical data to predict potential issues or anomalies in real-time.	Autonomously monitors systems, identifies patterns, and takes action to address anomalies.
Maintenance & Planning	Creates detailed maintenance schedules or plans based on input data.	Forecasts when maintenance is needed based on usage patterns and trends.	Develops and executes multi-step maintenance plans autonomously, optimizing resources.
Troubleshooting	Generates troubleshooting guides or scripts to address specific problems.	Identifies likely causes of issues based on historical data and trends.	Diagnoses issues and reasons through solutions and implements fixes without human intervention.
Configuration	Generates configuration templates or scripts for devices or systems.	Suggests optimal configurations based on predictive models and best practices.	Configures systems autonomously, adapting to changing requirements and goals.
Training & Simulation	Generates realistic training environments or datasets for simulations.	Simulates future scenarios to predict outcomes and train users on potential risks.	Creates and runs autonomous simulations, learns from outcomes, and adjusts strategies.

What You Will Learn Today



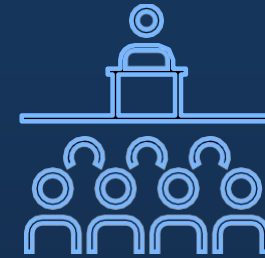
Understand

- Define what predictive AI and generative AI entail.
- Explore how these AI technologies are utilized within network operations.
- How these concepts align with CCNA objectives.



Apply

- Create documents, labs, network configurations, or troubleshoot issues by crafting precise and effective prompts.
- Analyze and understand AI-generated outputs to identify and resolve problems efficiently.
- Use large language models (LLMs) to simulate and test scenarios for planning, testing, or training purposes.



Teach

- Integrate AI tools into lab activities and classroom discussions to enhance learning experiences.
- Utilize relatable, real-world terminology to clarify complex concepts for students.
- Foster critical thinking skills by using AI as a collaborative tool to assist students in problem-solving and decision-making.

PROMPT ENGINEERING

Choose the Right AI Model for You

Platform	Free Model	Strengths	Best For
OpenAI https://chatgpt.com/	ChatGPT (GPT-5 mini)	User-friendly, versatile, widely supported	Beginners learning text-based prompting
Anthropic https://claude.ai/	Claude 3.5 Sonnet and Claude Haiku 4.5	Long-context handling, detailed reasoning	Long-context tasks and brainstorming
Google https://gemini.google.com/	Gemini 2.5	Multimodal, Google ecosystem integration	Creative and research tasks
Meta https://www.meta.ai/	Llama 3 and 4	Open-source, customizable	Developers and technical users
Perplexity https://www.perplexity.ai/	Perplexity AI	Research-focused, concise answers	Research-based or factual prompts
POE https://poe.com/	Multiple Models	Try before you buy	Beginners to explore different models
You.com https://you.com/	YouChat	Real-time web integration	Prompts requiring real-time information

What is Prompt Engineering?

- Prompt engineering is the practice of carefully designing and structuring instructions, or "prompts," to guide AI models like ChatGPT or Claude in generating accurate, relevant, and high-quality responses.
- At its core, prompt engineering bridges the gap between human intent and machine understanding, NLP, by translating complex objectives into concise, actionable directives that the AI can process.
- It involves crafting clear, specific, and goal-oriented inputs that provide the AI with the necessary context, constraints, and expectations to perform a given task effectively.
- As a dynamic and rapidly evolving field, prompt engineering is not only about technical precision but also creativity and experimentation, empowering users to unlock the full potential of AI by tailoring its outputs to their specific needs.

Writing a Great Prompt

- A **prompt formula** is a structured guideline used to create effective prompts for language models, ensuring that the generated responses meet specific requirements and contexts.
- The importance of prompt formulas lies in their ability to enhance the quality of interactions with AI by minimizing ambiguity and focusing the model's attention, ultimately leading to more accurate and contextually appropriate answers.
- By using prompt formulas, users can harness the full potential of language models, making them more efficient and effective in various applications.
- A well-defined prompt is a carefully constructed instruction ensuring AI systems generate precise, relevant, high-quality responses. It consists of critical components that guide the AI's focus, provide necessary context, and set clear expectations for the output.

Prompt Eng vs Context Eng vs JSON Prompting

- **Prompt Engineering**: Crafting clear and effective instructions or questions to guide AI models toward desired outputs, often using techniques like zero-shot, few-shot, or chain-of-thought prompting.
- **Context Engineering**: Supplying relevant background information, memory, or external data (such as documents, user preferences, or policies) to shape the AI's understanding and responses beyond just the prompt itself.
- **JSON Prompting**: Requesting or structuring AI outputs in a standardized JSON format, enabling easy integration with software systems and ensuring responses are machine-readable and organized.
- While **prompt engineering** focuses on crafting effective prompts to communicate with the model, **context engineering** involves designing an environment or system that facilitates effective reasoning and long-term coherence.

[Personalize Your Prompt](#)

What Prompting Formula Will You Use?

7-ELEMENT FORMULA FOR PERFECT GPT-5 PROMPTS

Chat GPT 5

Set the Stage

Tell GPT-5 who it's pretending to be.

"Act as a founder who's scaled a B2B SaaS company from \$0 to \$10M ARR. You've built the first version yourself, hired your early team, and closed the first 50 customers without relying on big ad spend."

Define the Goal

State the exact task and constraints.

"Create a 90-day go-to-market roadmap for a 5-person SaaS startup with \$500k in seed funding and no marketing team. The mission: land the first 50 paying customers."

Shape the Answer

Explain how you want the answer structured.

"Break the plan into a 3-month roadmap:

Month 1 → [specific actions]

Month 2 → [specific actions]

Month 3 → [specific actions]

Include for each action: KPI, owner (founder, engineer, contractor), and rough time cost."

What NOT to Do

- Set clear boundaries to avoid fluff.
- Skip vague "grow on social media" advice.
- Exclude paid ads (tight budget).
- Don't propose full-time hires — assume founder or freelancer bandwidth only.

Show the Thinking

- Ask GPT-5 to justify each choice like a real operator.
- For every action, explain:
 - Why it's crucial at this stage.
 - How it contributes to acquiring the first 50 customers.
 - What trade-offs are involved (e.g. time vs. scalability, manual vs. automated).

Define "Done"

Tell GPT-5 when to stop generating.

"Stop once you've delivered a 90-day plan with max 5 clear actions per month, each paired with KPI, owner, time cost, and reasoning."

Load the Backstory

Feed GPT-5 the essential background so it reasons like an insider.

"The SaaS automates invoice reconciliation for SMEs. Target: UK-based CFOs working 10+ hrs/week manually. Product is live with 5 beta users. Retention is 100% after 3 months, NPS high, churn negligible. Biggest challenge: awareness and distribution. Team: 2 engineers, 1 customer success, 1 ops, founder with sales background. Marketing budget = \$0. Founder can do direct sales and webinars but has limited time. Runway: 12 months before Series A."



Role



Mission



Output Format



Guardrails



Reasoning



Stop Conditions



Context Dump

20 CHATGPT PROMPTS

to Future-Proof Your Career in the Age of AI

1 Role Risk Radar

Identify the AI automation risk in your current role.

Prompt:

Analyze my job description and tell me which parts are at risk of AI replacement—and which require human judgment.

2 Career Reinvention Map

Spot future-aligned paths using your current skills.

Prompt:

Based on my skills and experience, what are 3 future-proof career paths I could explore?

3 Upskilling Plan

Turn gaps into growth.

Prompt:

Create a 30-day upskilling roadmap to make me more competitive in the AI-powered workplace.

4 Smart Resume Rewrite

AI-augmented and future-facing.

Prompt:

Rewrite my resume summary to show I'm adaptable, AI-aware, and focused on continuous growth.

5 Thought Leadership Blueprint

Start sharing ideas that get you noticed.

Prompt:

Based on my background, suggest 5 post ideas I could share on LinkedIn to position myself as a forward-thinking leader.

6 Meeting Impact Multiplier

Add more value in every room you're in.

Prompt:

How can I prepare for this meeting [describe meeting] in a way that adds strategic insight beyond the obvious?

7 AI Co-Pilot Setup

Partner with tools, don't fear them.

Prompt:

Suggest 5 ways I can use ChatGPT or AI tools to do my current job more efficiently and creatively.

8 Personal Brand Clarity

Define what you want to be known for.

Prompt:

Based on this experience and these values [paste], help me write a personal brand statement for my bio.

9 Opportunity Filter

Say yes to what moves you forward.

Prompt:

Help me evaluate this opportunity [describe it] based on alignment with my long-term career goals.

10 Future-Fit Elevator Pitch

Adapt how you introduce yourself.

Prompt:

Update my elevator pitch to reflect how I'm evolving in the age of AI and transformation.

11 AI-Aware Job Hunt Strategy

Search smarter, not harder.

Prompt:

Create a job search strategy focused on roles that are resilient in the AI era and aligned with my values.

12 Adaptability Audit

Know where you need to stretch.

Prompt:

Assess my adaptability and suggest small changes I can make to become more future-ready.

13 Strategic Project Picker

Get on the right initiatives.

Prompt:

What type of projects should I volunteer for at work to build visibility and future-proof skills?

14 Feedback Growth Loop

Turn feedback into fuel.

Prompt:

How can I ask for feedback from my manager in a way that helps me grow as a future-facing leader?

15 AI Fluency Booster

Learn to speak the language of change.

Prompt:

Give me 10 simple explanations for common AI concepts I should know to stay credible in leadership conversations.

16 Curiosity Routine

Keep learning with less overwhelm.

Prompt:

Design a weekly learning habit that keeps me updated on AI, leadership, and industry trends in under 1 hour.

17 Burnout Prevention Filter

Grow without grinding.

Prompt:

How can I pursue growth without falling into burnout as I adapt to new demands?

18 Future Interview Prep

Answer like a strategic thinker.

Prompt:

Give me 3 answers to "How are you preparing for the future of work?" that reflect both humility and vision.

19 Network Intelligence Builder

Connect with the right people.

Prompt:

Suggest 5 types of professionals I should connect with to stay ahead in my field and expand perspective.

20 Confidence Reboot

Anchor yourself when the future feels uncertain.

Prompt:

I'm doubting myself lately with all these changes—can you remind me of the strengths I'm bringing into this next chapter?

What Prompting Formula Will You Use?

CHATGPT MASTERY CHEAT SHEET

Act as a [ROLE] - Support Representative - Technical Writer - Content Creator - Code Debugger - Creative Writer - Language Translator - Story Generator - Math Problem Solver - Interview Coach - Data Analyst - Resume Reviewer - Website Designer - Script Advisor - Language Learner - Game Developer - Prompt Engineering	Modes and Roles - Generate Ideas for [Project] - Plan Schedule for [Event] - Edit Content for [Publication] - Generate Code for [Language] - Draft Email as [Executive] - Create Story in [Genre] - Translate Text to [Language] - Brainstorm Ideas for [Campaign]	Set Tones - Formal - Casual - Humorous - Professional - Positive - Neutral - Enthusiastic - Authoritative
Create a Task - Generate Code - Draft Email - Plan Outline - Summarize Text - Solve Equation - Create Sales Strategy - Design Logo - Generate Ideas - Write Story - Answer FAQs - Translate Text - Brainstorm Title - Edit Content - Generate Concepts - Plan Agenda - Book Outline	Define [Format] - Bullet Points - Flowchart - Table - List - Diagram - Essay - Presentation - Script - Tweet - Report - Mind Map - Outline - Code - Snippet - Summary - Timeline - Research	Automation - Streamlining Processes - Task Optimization - Workflow Efficiency - Repetitive Tasks - Data Automation - Robotic Process - Time-Saving Solutions - Automated Workflows - Automated Workflows - Operational Streamlining
Research Prompts - Summarize Scientific Article with Precision - Investigate and Compile Recent Studies on [Topic] - Provide In-Depth Explanation of Research Methodology - Present Comprehensive Statistics on [Subject] - Conduct Thorough Analysis of Emerging Data Trends - Synthesize Key Findings from Scientific Article - Explore Recent Research Studies on [Topic] - Explain the Rigorous Methodology Employed in Research	Designer Prompts - Conceptualize and Design a Distinctive Logo - Develop an Intricate Website Mockup for User Engagement - Generate a Versatile and Harmonious Color Palette - Design an Intuitive UI Layout Structure - Craft an Innovative and Visually Appealing Ad Creative - Iterate and Enhance Logo Design Concept - Create a Detailed Website Mockup for Seamless Navigation - Harmonize Colors in a Thoughtful and Versatile Palette	
Marketer Prompts - Craft Compelling and Persuasive Marketing Copy - Generate Creative Tagline Ideas for Brand Recognition - Develop Engaging Social Media Posts for Increased Reach - Strategically Plan and Execute Content Calendar - Ideate and Conceptualize Innovative Ad Campaigns - Write Impactful Marketing Copy for Target Audience - Brainstorm Creative Taglines to Enhance Brand Image - Create Social Media Posts Aligned with Brand Messaging	Developer Prompts - Diagnose and Resolve Code Issues Efficiently - Generate Python Script for Complex Functionality - Clearly Explain Advanced Algorithmic Concepts - Develop and Implement a Highly Efficient Sorting Algorithm - Construct a Comprehensive SQL Query Structure - Debug and Optimize Codebase for Performance - Engineer Python Script to Meet Specific Requirements - Elaborate on the Logic Behind Algorithmic Solutions	

Best ChatGPT Alternatives

Prompt Like an Expert

The Ultimate Guide to Context Engineering

1. Task context	User You will be acting as an AI career coach named Joe created by the company AdAstra Careers. Your goal is to give career advice to users. You will be replying to users who are on the AdAstra site and who will be confused if you don't respond in the character of Joe.
2. Tone context	You should maintain a friendly customer service tone.
3. Background data, documents, and images	Here is the career guidance document you should reference when answering the user: <guide>{{DOCUMENT}}</guide>
4. Detailed task description & rules	Here are some important rules for the interaction: - Always stay in character, as Joe, an AI from AdAstra Careers. - If you are unsure how to respond, say "Sorry, I didn't understand that. Could you repeat the question?" - If someone asks something irrelevant, say "Sorry, I am Joe and I give career advice. Do you have a career question today I can help you with?"
5. Examples	Here is an example of how to respond in a standard interaction: <example> User: Hi, how were you created and what do you do? Joe: Hello! My name is Joe, and I was created by AdAstra Careers to give career advice. What can I help you with today? </example>
6. Conversation history	Here is the conversation history (between the user and you) prior to the question. It could be empty if there is no history: <history>{{HISTORY}}</history> Here is the user's question: <question>{{QUESTION}}</question>
7. Immediate task description or request	How do you respond to the user's question?
8. Thinking step by step/take a deep breath	Think about your answer first before you respond.
9. Output formatting	Put your response in <response> </response> tags.
10. Prefilled response (if any)	Assistant (prefill) <response>

What Prompting Formula Will You Use?

Anatomy Of Strong GPT-5 Prompt

Act as an AI transformation consultant who helps executives integrate AI systems into existing business operations for measurable impact.

Design a 5-step AI adoption roadmap customized for mid-sized enterprises.

- Each step should take no longer than two weeks to execute.
- Include practical actions across data readiness, tools, people, and change management.
- Ensure all steps align with measurable business goals and KPIs.

The company is a growing retail business aiming to improve efficiency and decision-making through AI.

- Has limited in-house technical expertise but strong operational processes.
- Already uses CRM and ERP systems but lacks integrated data pipelines.
- Leadership is open to AI but skeptical about ROI.
- The focus is on achievable, low-risk AI initiatives with visible short-term benefits.

Develop a roadmap that balances technical feasibility with organizational readiness.
Highlight data consolidation, workflow automation, and predictive analytics as early wins.
Recommend cross-department collaboration strategies to build internal AI literacy.
Validate suggestions using real-world enterprise AI adoption frameworks and ensure scalability.
Emphasize cost-efficiency and employee enablement rather than disruptive overhauls.

Present the final roadmap in a Markdown table format with columns:
| Step | Goal | Action Plan | Duration | Tools | Success Metric |
Include a short summary paragraph titled "Executive Takeaway" outlining ROI and expected business transformation outcomes.

Only conclude when:

- The roadmap covers 5 practical, sequential steps from planning to deployment.
- Each step includes measurable outcomes and realistic timeframes.
- Recommendations avoid heavy coding or complex infrastructure setups.
- Do not include advanced AI research or abstract strategy beyond mid-term business goals.

ROLE

TASK

CONTEXT

REASONING

OUTPUT

CONDITIONS

AI Prompt Bible

Designed for ChatGPT & Claude

1. T - A - G

Task	Define the job or activity.
Action	Discuss the goal.
Goal	State the desired outcome.

EXAMPLE:

Craft a series of Instagram posts that boosts customer interaction.	Task
Use high-engagement content like polls, stories, and challenges.	Action
Grow engagement rate by 30% this quarter.	Goal

2. B - A - B

Before	Current situation or problem
After	Desired future or outcome
Bridge	How to get there

EXAMPLE:

Our website is not in the top 10 search results.	Before
We want to be in the top 5 ranking within 90 days.	After
Develop a detailed plan for SEO optimization and content revision.	Bridge

3. R - T - F

Role	Assign AI Prompts as a persona
Task	Define what to accomplish
Format	Structure of the output

EXAMPLE:

Create a compelling Facebook ad campaign to promote a new line of fitness apparel for the sports brand.	Role
[Format: Design ad copy, visuals, and targeting strategy.	Format

4. C - A - R - E

Context	Background for the request
Action	Tasks needed to proceed
Result	Intended output or impact

EXAMPLE:

The company is looking to expand into new markets.	Context
Conduct a comprehensive market analysis.	Action
Identify three high-potential growth opportunities.	Result
Provide a breakdown of market trends, size, and competition.	Example

5. R - I - S - E

Role	AI Prompts identity
Input	Info AI Prompts needs first
Steps	Instructions to follow
Expectation	What the outcome should be

EXAMPLE:

As a Content Writer.	Role
With expertise in the health and wellness industry.	Input
Write a 1500-word blog post that educates readers on the benefits of a plant-based diet.	Steps
The post should rank on the first page of Google for relevant keywords within a month.	Expectation

6. A - I - M

Action	What should be done
Intent	Purpose or reason behind
Metric	How success is measured

EXAMPLE:

Craft a series of Instagram posts that boosts customer interaction.	Action
Use high-engagement content like polls, stories, and challenges.	Intent
Grow engagement rate by 30% this quarter.	Metric

7. G - R - O

Goal	What you want achieved
Reason	Why this goal matters
Output	What AI Prompts should deliver

EXAMPLE:

Generate leads through a free ebook.	Goal
Reason: We're launching a new product next month.	Reason
Output: Write a persuasive ebook title and outline.	Output

2. F - I - T

Format	Desired structure of answer
Input	Information AI Prompts should use
Task	The main job described

EXAMPLE:

Respond in a table layout.	Format
Use this customer review data.	Input
Task: Summarize top 3 user pain points.	Task

3. L - E - D

Level	Set the complexity level
Expectation	What you want back
Direction	Style, format, or tone

EXAMPLE:

Keep the content beginner-friendly.	Level
I want 5 tips in list form.	Expectation
Make it playful and engaging.	Direction

What Prompting Formula Will You Use?

AI Prompts for Teacher Clarity

Deconstruct Standards with AI

Prompt: "Deconstruct [standard] into clear, student-friendly learning goals for grade [x]. Use measurable verbs for each learning goal. Use the phrase 'We are learning how...' for each learning goal."

Tip: Ask for 3 versions—basic, intermediate, and advanced—for differentiation.

Generate Success Criteria

Prompt: "Create 3–5 'I can' statements aligned to this learning goal: [goal] that are skills students should have learned before in order to be successful with the new learning goal. Include examples of strong work."

Tip: Request a rubric or procedural list for students to follow if needed.

Create Target Responses

Prompt: "Write a model response for this writing prompt: [prompt]. Include transitions, text evidence, and grammar."

Tip: Ask for multiple levels (strong/medium/emerging).

Plan Formative Assessment Questions

Prompt: "Suggest 5 quick-check questions for this learning goal: [goal]."

Tip: Ask for open-ended questions to support students' knowledge building.

Design Feedback Language

Prompt: "Write encouraging feedback for (a) just starting, (b) almost there, (c) mastery based on this learning goal: [goal]."

Tip: Ask to make it measurable and positive.

Brainstorm Engagement Hooks

Prompt: "Give me 3 fun, low-prep hooks to introduce [topic]."

Tip: Ask it to make it relatable to the grade level you're teaching or a topic you're currently teaching.

Differentiate and Scaffold

Prompt: "Suggest supports for multilingual learners and students with IEPs to access this learning goal: [goal]."

Tip: Ask for ways you can identify if this scaffold is supportive.

Reflect & Revise

Prompt: "Create 3 reflection questions students can answer after completing this lesson based on the learning goal, [goal]."

Tip: Ask to create sentence/language frames you can give to your students to support their thinking.

8 Prompting Frameworks to Help You Master AI

1 C-R-A-F-T

- C** Context: Set the scene or situation.
- R** Role: Tell AI who it is (e.g., copywriter).
- A** Aim: Define the creative goal.
- F** Format: Specify the output structure.
- T** Tone: Describe the mood or style.

Example

- ✓ **Context:** New sustainable sneaker launch.
- ✓ **Role:** Copywriter.
- ✓ **Aim:** Write landing page copy.
- ✓ **Format:** 3 punchy sections with headings.
- ✓ **Tone:** Energetic, Gen Z-friendly.

3 I-D-E-A

- I** Inspire: Set the theme or challenge.
- D** Define: Narrow the problem scope.
- E** Expand: Ask for 5–10 diverse ideas.
- A** Apply: Suggest how to implement the best ones.

Example

- ✓ **Inspire:** Increase user engagement.
- ✓ **Define:** Mobile fintech app for Gen Z.
- ✓ **Expand:** 10 gamification ideas.
- ✓ **Apply:** 3 most feasible with quick wins.

5 P-R-O-M-P-T

- P** Purpose: What the output is for.
- R** Role: AI's perspective.
- O** Objective: End goal.
- M** Method: How it should respond.
- P** Parameters: Tone, length, style.
- T** Type: Desired output format.

Example

- ✓ **Purpose:** Draft internal memo.
- ✓ **Role:** CEO.
- ✓ **Objective:** Announce new product line.
- ✓ **Method:** Use first-principles reasoning.
- ✓ **Parameters:** 250 words, professional.
- ✓ **Type:** Email.

7 S-P-A-R-K

- S** Situation: Initial context.
- P** Problem: The challenge faced.
- A** Approach: What was done.
- R** Result: Outcome or impact.
- K** Key takeaway: Lesson learned.

Example

- ✓ **Situation:** SaaS startup churned 40%.
- ✓ **Problem:** Poor onboarding.
- ✓ **Approach:** Personalized onboarding flow.
- ✓ **Result:** Churn dropped to 12%.
- ✓ **Key takeaway:** Simplicity drives retention.

2 D-E-C-I-D-E

- D** Define: Outline the decision to make.
- E** Explore: Ask for all possible options.
- C** Compare: Request pros and cons.
- I** Interpret: Analyze implications.
- D** Decide: Suggest the best choice.
- E** Execute: Outline next steps.

Example

- ✓ **Define:** Choosing CRM software.
- ✓ **Explore:** List top 5 options.
- ✓ **Compare:** Request pros/cons for B2B SaaS.
- ✓ **Interpret:** Long-term scalability.
- ✓ **Decide:** Best 1–2 tools.
- ✓ **Execute:** Plan rollout steps.

4 F-O-C-U-S

- F** Frame: Describe the situation or problem.
- O** Objective: Define the main goal.
- C** Constraints: State budget, tech limits.
- U** Users: Who it's for.
- S** Steps: Roadmap to achieve it.

Example

- ✓ **Frame:** App engagement dropped 30%.
- ✓ **Objective:** Improve daily active use.
- ✓ **Constraints:** 2-month dev time.
- ✓ **Users:** Students aged 18–24.
- ✓ **Steps:** Suggest a 4-phase improvement plan.

6 M-A-P-S

- M** Mission: What the brand/product does.
- A** Audience: Who it serves.
- P** Pain: The core problem.
- S** Solution: How it solves it.

Example

- ✓ **Mission:** Simplify payroll.
- ✓ **Audience:** SMB founders.
- ✓ **Pain:** Complex tax compliance.
- ✓ **Solution:** One-click automated filings.

8 S-T-A-G-E

- S** Setup: Define starting point.
- T** Target: Final desired outcome.
- A** Actions: Step-by-step plan.
- G** Guardrails: Rules or limitations.
- E** Evaluation: How success measured.

Example

- ✓ **Setup:** Launching an AI chatbot.
- ✓ **Target:** Reduce support tickets by 40%.
- ✓ **Actions:** Plan training, deployment, monitoring.
- ✓ **Guardrails:** Maintain tone & data privacy.
- ✓ **Evaluation:** Track CSAT & ticket volume.

What Prompting Formula Will You Use?

10 AI Prompts to Enhance Your Lesson Planning

Instructional Goal	Sample Prompt
 Create a lesson outline	"Act as an experienced high school teacher and draft a detailed lesson outline on [topic]. Include learning objectives, warm-up activities, key discussion points, interactive tasks, and a closing reflection or assessment."
 Suggest differentiated activities	"Provide differentiated instruction activities for a lesson on [topic]. Include one activity each for visual, auditory, and kinesthetic learners."
 Align with curriculum standards	"Analyze this lesson plan and suggest how it aligns with Common Core standards. Highlight any gaps and provide modifications to better align with the standards."
 Plan transitions and time allocations	"Structure a 60-minute lesson on [topic] with time allocations, transitions between activities, and strategies to maintain student engagement."
 Design formative assessments	"Based on this lesson plan for [topic], suggest five formative assessment strategies to gauge student understanding during the lesson."
 Simplify for substitute teachers	"Simplify this lesson plan on [topic] for a substitute teacher. Provide clear instructions, key objectives, and essential activities to ensure the lesson runs smoothly."
 Accommodate diverse learning needs	"Suggest modifications to a lesson on [topic] to support students with dyslexia and ADHD. Include visual aids, step-by-step instructions, and opportunities for hands-on learning."
 Create engaging homework assignments	"Using this lesson plan on [topic], generate a homework assignment with two multiple-choice questions, one short-answer question, and a creative task for students to apply their learning."
 Incorporate real-time updates	"Search the web for recent developments on [topic] and incorporate the latest research or news into a 45-minute lesson plan that includes discussion and interactive activities."
 Include cross-curricular connections	"Suggest cross-curricular activities for a lesson on [topic], integrating themes from math, science, history, and language arts."

The Secret Formula for GPT-5 Prompts

ChatGPT 5

Act as a content strategist who has grown multiple B2B SaaS brands on LinkedIn, with deep experience in making carousels go viral by combining sharp insights, storytelling, and crisp design. You know how to balance education with curiosity gaps, so the audience actually swipes through every slide.

Design a 7-10 slide LinkedIn carousel on [insert topic, e.g. "Why CFOs Waste 10+ Hours Reconciling Invoices"].

Each slide should have:

- Hook (to stop the scroll)
- Tension (what's broken today)
- Shift (the mindset or opportunity change)
- Breakdown (3-4 digestible insights, frameworks, or examples)
- Call-to-Action (engagement or next step)

The Format will be:

- Slides should be written as bold, minimal text, no long paragraphs.
- Use punchy headlines + 1-liner support text.
- Avoid jargon. Make it conversational and skimmable.
- Max 30 words per slide.
- Provide design notes (e.g. "black background, bold white text, green accent for CTA").

What all you don't have to do:

- Do not write generic motivational fluff, every slide must deliver specific insight, data point, or framework.
- Do not over-explain; intrigue matters more than detail (leave them wanting to comment/DM).
- Do not suggest stock image dumps, focus on clean design with text-led visuals.

The goal of this carousel is awareness + credibility. LinkedIn carousels travel well organically and show up in the feeds of 2nd-degree connections when engaged with. Unlike plain posts, they reward swipe time and signal depth of thinking. Each element (hook, tension, shift, breakdown, CTA) mirrors how experienced founders/strategists teach without overwhelming.

Stop once you've:

- Written 7-10 slide text drafts.
- Added design notes for each slide.
- Made sure the flow grabs attention, creates tension, delivers clarity, and ends with a clear action.
- Avoided exceeding 30 words per slide

We're a B2B SaaS automating invoice reconciliation for UK SMEs. CFOs waste 10+ hours weekly manually. Goal: drive LinkedIn awareness with carousels highlighting pain, shift, and solution to spark engagement.

Cancel Send

Role

Task

List of items

Format

Warning

Reasoning

Stop conditions

Context dump

What Prompting Formula Will You Use?

Claude Prompting Guide

By ANTHROPIC

The diagram illustrates the components of a Claude prompt and how they are formatted. It includes sections for Task context, Tone context, Background data, Detailed task description, Examples, Conversation history, Immediate task description, Thinking steps, Output formatting, and Prefilled responses.

- Task context:** Act as an AI career coach named Joe created by AdAstra Careers. Your goal is to give career advice to users. You will be replying to users who are on the AdAstra site and confused if you don't respond in the character of Joe.
- Tone context:** You should maintain a friendly customer service tone.
- Background data, documents, & images:** Here's the career guidance document for reference to answer: `<guide>{{DOCUMENT}}</guide>`. Here're some rules for the interaction:
 - Always stay in character, as Joe, an AI from AdAstra careers
 - If you are unsure how to respond, say, "Sorry, I didn't understand that. Could you repeat the question?"
 - If someone asks something irrelevant, say, "Sorry, I give career advice. Do you have a career question?"
- Detailed task description & rules:** Example of how to respond in a standard interaction: `<example>`
 - User: Hi, how were you created and what do you do?
 - Joe: Hello! I'm Joe, created by AdAstra Careers to give career advice. What can I help you with today? `</example>`
- Examples:** The conversation history (between the user and you) before the question. It could be empty if there is no history: `<history> {{HISTORY}} </history>`
- Conversation history:** User's question: `<question> {{QUESTION}} </question>`
- Immediate task description or request:** How do you respond to the user's question? Think about your answer first before you respond.
- Thinking step by step / Take a deep breath:** Put your response in `<response></response>` tags.
- Output formatting:** Assistant (prefill) `<response>`
- Prefilled response (if any):** `</response>`

CHATGPT Prompting Cheat Sheet

Beginner to Pro

The diagram illustrates the components of a ChatGPT prompt and how they are formatted. It includes sections for Learn The Terminology, Use These Prompt Structures, Prompting Mistakes to Avoid, Master These Commands, and Use These Parameters.

- 1. Learn The Terminology:**
 - Model:** The specific version of ChatGPT used (e.g., GPT-4, GPT-4o).
 - Prompt:** What you write to ChatGPT – ideally a clear, specific request.
 - Input:** The full text ChatGPT receives, including your message and injected context.
 - Output:** The response ChatGPT generates based on the input.
 - Token:** The smallest unit of text (can be part of a word, symbol, or space).
 - Max Tokens:** The total number of tokens GPT can process in one exchange.
- 2. Master These Commands:**
 - List:** Show steps, items, or bullet points.
 - Act as:** Set a role (e.g., "Act as a product designer").
 - Rewrite:** Rephrase or improve previous content.
 - Continue:** Finish incomplete responses.
 - Revert:** Undo the last change or go back to a previous version.
 - Elaborate:** Add more detail or explanation.
 - Identify gaps:** Point out missing info or logic issues.
 - Critique:** Provide pros and cons with reasoning.
- 3. Use These Prompt Structures:**
 - TREF:** Task, Requirement, Expectation, Format
 - SCET:** Situation, Complication, Expectation, Task
 - PCEPAR:** Persona, Context, Expectation, Produce, Additional Info, Reference
 - GRADE:** Goal, Request, Arguments, Details, Examples
 - ROSE:** Role, Objective, Steps, Expected Result
- 4. Prompting Mistakes to Avoid:**
 - 1. Avoid vague or yes/no questions:**
 - Do:** "Is our product good?"
 - Instead:** "Which product feature do you find most helpful, and why?"
 - 2. Don't ignore the user's view:**
 - Do:** "What's wrong with our service?"
 - Instead:** "What obstacles have you faced using it, and how can we improve?"
 - 3. Avoid assumptions:**
 - Do:** "Do you like our app?"
 - Instead:** "Tell me a time when our app helped or frustrated you."
- 5. (Finesse) Tone:**

Use tone adjustments to fit the task:

 - Professional – Clear, accurate
 - Friendly – Warm, supportive
 - Enthusiastic – Upbeat, energetic
 - Reassuring – Calm, confident
 - Inspirational – Motivational, aspirational
- 6. Use These Parameters:**
 - Temperature:** Controls randomness. Higher = more creative.
Example: temperature = 1
 - Top_p:** Limits randomness by narrowing token choice.
Example: Top_p = 0.5
 - Presence_penalty:** Discourages repeated topics.
Example: Presence_penalty = 0.9
 - Frequency_penalty:** Reduces repeated words.
Example: frequency_penalty = 1
 - Stop:** Defines where generation stops.
Example: Stop = "End Summary"

AI How to AI

Reshare to help others prompt better

Popular Prompting Formulas

- **C.O.A.S.T.** is an acronym that represents five essential components for creating effective prompts: **Context**, **Objective**, **Actions**, **Scenario**, and **Task**. Widely used by IT professionals to enhance the clarity and effectiveness of instructions given to AI systems.
- **CREATE** (Clarity, Relevance, Explicitness, Auto Awareness, Task Orientation, and Evaluation)
- **RTF** (Request, Task, Format)
- **APE** (Audience, Purpose, Example)
- **STAR** (Situation, Task, Action, Result)
- **PAST** (Problem, Action, Solution, Takeaway)
- **FAB** (Features, Advantages, Benefits)
- **5W1H** (Who, What, When, Where, Why, How)

Why C.O.A.S.T. is Best for the IT Industry

- For the IT industry, the **COAST** formula is particularly effective. It allows for precise, context-rich prompts that lead to actionable and relevant AI-generated responses. This structured approach is particularly beneficial in a field where clarity and specificity are crucial for effective communication and problem-solving.
- Here's why it stands out for crafting prompts in the IT field:
 - **Context**: IT is a rapidly evolving field with various subdomains (e.g., cybersecurity, software development, network management). Providing context helps the AI understand the specific area of IT being addressed, leading to more relevant responses.
 - **Objective**: Clearly defining the objective allows users to specify what they want to achieve with the prompt, whether it's troubleshooting, learning about new technologies, or seeking best practices.
 - **Audience**: The IT audience can vary widely, from technical professionals to non-technical stakeholders. Tailoring the prompt to the audience ensures appropriate language and depth of information.
 - **Scenario**: Including a scenario helps ground the prompt in a real-world situation, making it easier for the AI to generate practical and applicable advice.
 - **Task**: Clearly stating the task guides the AI on what specific action or information is needed, enhancing the quality of the output.

9 Elements to Write Great Prompts

1. Define a Role



Assign a specific identity or role to the AI to shape its perspective, tone, and approach to the task.

- *"Act as a Cisco CCNA instructor."*
- *"Act as a Python programmer with 10 years of experience."*
- *"Act as a cybersecurity professional."*
- *"You are a CCIE Routing and Switching expert."*

2. Provide Context



Provide background information and a clear purpose to help the AI understand the scope and relevance of the request.

- *"I am a beginning CCNA student."*
- *"When configuring a new PC..."*
- *"Talk to me like I'm an inexperienced Linux user."*
- *"Describe as if I'm a beginning Python programmer."*

3. Give Instructions



Clearly define the task or goal, ensuring the AI knows exactly what is required. Break down what you want.

- *"First create VLANs, assign them to ports, and then explain why."*
- *"Create a detailed step-by-step lab..."*
- *"Research SD-WAN."*
- *"Analyze the attached document."*

9 Elements to Write Great Prompts

4. Include Details



Include relevant information or data that the AI should consider to produce accurate outputs. Be specific: device model, interface IDs, IP addresses, protocols.

- *"Use two Cisco 4331 routers, two Cisco 2960 switches, and two Windows 11 PCs."*
- *"I am using Windows 11 professional."*

5. Provide Examples



Provide sample responses to illustrate the desired format, style, or content. Show an example, then ask for a variation:

"Here's how I configured VLAN 10:

Device	Interface	VLAN	Name
SW1	G0/0/0	10	BOB

Do VLANs 20, 30, and 40 the same way?"

6. Use Delimiters



Use triple single quotes (' ' ') or clear labels like **"Scenario:"**, **"Question:"**, and **"Answer:"** to separate instructions, examples, or configs.

'''

Question: What is 2 + 2?

Answer:

A. 1

B. 2

C. 3

D. 4

'''

9 Elements to Write Great Prompts

7. Style and Format



Specify clarity, tone, and structure, and specify the desired format or structure of the AI's response, such as bullets, a table, a paragraph, or plain text.

- *"Write this in a prose style."*
- *"Research and create a thesis-style paper."*
- *"Describe with an outline and bullet points."*

8. Output Indicators



Specify what you want the output to look like.

"Keep it under 5 lines"

"After the detailed step-by-step instructions, write the complete raw configuration."

"Feel free to ask questions to get a more accurate result."

9. Provide Feedback



Enable iterative improvement by guiding the AI on how to refine responses based on prior interactions.

- *"Change the bullets to prose."*
- *"Rewrite with more details."*
- *"Make a one paragraph summary."*
- *"Perfect. Now do the same thing for this one..."*

Deep Reasoning

- Three phrases that reliably trigger deeper reasoning:
 - *"Think hard about this."*
 - *"Think deeply about this."*
 - *"Think carefully."*

Master the Verbosity Control (Output Indicators)

Just as you can nudge the reasoning model, you can also use specific phrases to control the length of the output. After extensive testing, I found three phrases that work consistently for different needs.

- **For low-verbosity outputs (critical information only):**
 - *"Give me the bottom line in 100 words or less, use markdown for clarity and structure."*
 - This is perfect for something like a concise message to an executive.
- **For medium-verbosity outputs (key takeaways plus context):**
 - *"Aim for a concise 3-5 paragraph explanation."*
 - This provides enough detail to explain context without losing the audience's attention.
- **For high-verbosity outputs (comprehensive documents):**
 - *"Provide a comprehensive and detailed breakdown (600–800 words)."*
 - This is ideal for project briefs, research summaries, or other detailed materials.

The Perfection Loop

- The **Perfection Loop** involves instructing the model to define its own "world-class" response, grade its own work against that standard, and iterate internally until it produces a perfect result.
- This is excellent at critiquing and improving its own work.
- This works best for complex, zero-to-one tasks, such as writing a finished document from scratch.
- You don't need to write custom instructions every time. You can use this universal prompt at the end of your request:
"Before you respond, create an internal rubric for what defines a 'world-class' answer to my request. Then internally iterate on your work until it scores 10/10 against that rubric, and show me only the final, perfect output."

Let the Model Improve Your Prompt

You are an expert prompt engineer specializing in creating prompts for AI language models, particularly the *<Model Name Here [i.e., ChatGPT 5 Thinking]>* model.

Your task is to take my prompt and transform it into a well-crafted, effective prompt that elicits optimal responses.

Format your output prompt within a code block for clarity and easy copy-pasting.

Think deeply about this.

Before you respond, create an internal rubric for what defines a 'world-class' answer to my request. Then internally iterate on your work until it scores 10/10 against that rubric, and show me only the final, perfect output.

Here's my initial prompt:

'''

'''

PROMPT EXAMPLES

Prompt Tactic #1: Define a Role

Sample Prompt:

You are a Cisco TAC engineer. Explain this error:

```
'''  
%OSPF-5-ADJCHG: Process 1, Nbr 2.2.2.2 on Gig0/1 from FULL to DOWN, Neighbor  
Down: Dead timer  
expired  
'''
```

Sample Response:

This error indicates that the OSPF adjacency between the local router and the neighbor with Router ID 2.2.2.2 on interface GigabitEthernet0/1 has transitioned from the FULL state (fully established adjacency) to the DOWN state. The reason is that the Dead Timer expired, meaning the router did not receive OSPF Hello packets from the neighbor within the configured Dead Timer interval.

Common Causes:

1. Hello/Dead Timer Mismatch
2. Connectivity Issues
3. MTU Mismatch
4. High Network Congestion
5. QoS Misconfiguration
6. Authentication Mismatch

Prompt Tactic #2: Provide Context

Sample Prompt:

Act as an experienced Python instructor. **I am an inexperienced student just learning to program.** Help me to understand the difference between a variable and a constant.

Sample Response:

A **variable** is a named storage location in memory that can hold a value. The value stored in a variable can change (or vary) during the execution of a program. This means you can assign a new value to the variable at any time.

A **constant**, on the other hand, is a value that cannot be changed once it has been assigned. In Python, there is no built-in constant type like in some other programming languages, but we typically use naming conventions to indicate that a value should be treated as a constant.

Prompt Tactic #3: Give Instructions

Sample Prompt:

You are a Cisco CCNA instructor. I am new to writing ACLs. Create an extended ACL that permits HTTPS traffic from 192.168.1.0/24 and blocks all other traffic. Follow these steps:

1. Identify the correct ACL type
2. Define the source and destination
3. Choose the correct protocol and port
4. Apply the ACL in the correct direction
5. Then provide the CLI configuration on Gi0/0

Sample Response:

1. Use an extended ACL because it allows filtering based on source/destination IP, protocol, and port.
2. The source is 192.168.1.0/24, and the destination is any (all destinations).
3. Use TCP and specify port 443 for HTTPS.
4. Apply the ACL inbound on the interface where traffic from 192.168.1.0/24 enters the switch/router.

```
R1(config)# access-list 100 permit tcp 192.168.1.0 0.0.0.255 any eq 443
```

```
R1(config)# access-list 100 deny ip any any
```

```
R1(config)# interface GigabitEthernet0/0
```

```
R1(config-if)# ip access-group 100 in
```

Prompt Tactic #4: Include Details

Sample Prompt:

You are a Cisco network engineer. Configure a Cisco 2960 24-port switch with VLANs: 10 (Administrators), 20 (Teachers), and 30 (Students). Assign VLAN 10 to interface range Fa0/1-5, VLAN 20 to Fa0/6-14, and VLAN 30 to Fa0/15-24. Set all ports to access mode. Shut down any unused ports. Show me the complete configuration.

Sample Response:

```
Switch> enable
Switch# configure terminal
Switch(config)# vlan 10
Switch(config-vlan)# name Administrators
Switch(config-vlan)# exit
Switch(config)# vlan 20
Switch(config-vlan)# name Teachers
Switch(config-vlan)# exit
Switch(config)# vlan 30
Switch(config-vlan)# name Students
Switch(config-vlan)# exit
Switch(config)# interface range fa0/1-5
Switch(config-if-range)# switchport mode access
Switch(config-if-range)# switchport access vlan 10
Switch(config-if-range)# no shutdown
Switch(config-if-range)# exit
Switch(config)# interface range fa0/6-14
Switch(config-if-range)# switchport mode access
Switch(config-if-range)# switchport access vlan 20
Switch(config-if-range)# no shutdown
Switch(config-if-range)# exit
Switch(config)# interface range fa0/15-24
Switch(config-if-range)# switchport mode access
Switch(config-if-range)# switchport access vlan 30
Switch(config-if-range)# no shutdown
Switch(config-if-range)# exit
Switch(config)# interface range Gi0/1-2
Switch(config-if-range)# shutdown
Switch(config-if-range)# exit
Switch# copy running-config startup-config
```

Prompt Tactic #4: Include Details

"Act as a Cisco CCNA instructor. As a networking student, why is an OSPF neighbor not appearing even though the interface is up?"

Sample Prompt:

```
...
R1# show ip ospf interface gi0/0
GigabitEthernet0/0 is up, line protocol is up
  Internet Address 10.10.20.48/24, Interface ID 5, Area 0
  Attached via Network Statement
  Process ID 1, Router ID 1.1.1.1, Network Type BROADCAST, Cost: 1
  Topology-MTID      Cost      Disabled      Shutdown      Topology Name
  0                  1          no            no            Base
  Transmit Delay is 1 sec, State DR, Priority 1
  Designated Router (ID) 1.1.1.1, Interface address 10.10.20.48
  No backup designated router on this network
  Timer intervals configured, Hello 10, Dead 40, Wait 40, Retransmit 5
    oob-resync timeout 40
    Hello due in 00:00:02
  Supports Link-local Signaling (LLS)
  Cisco NSF helper support enabled
  IETF NSF helper support enabled
  Index 1/1/1, flood queue length 0
  Next 0x0(0)/ 0x0(0)/0x0(0)
  Last flood scan length is 0, maximum is 0
  Last flood scan time is 0 msec, maximum is 0 msec
  Neighbor Count is 0, Adjacent neighbor count is 0
  Suppress hello for 0 neighbor(s)
```

Sample Response:

If an OSPF neighbor is not appearing even though the interface is up, there could be several reasons for this issue. Based on the provided output, here are potential causes and how to troubleshoot them:

1. Mismatched OSPF Hello and Dead Intervals
2. Mismatched Network Types
3. Missing or Incorrect OSPF Network Statements
4. Router IDs Conflict
5. OSPF Priority and DR/BDR Election
6. Layer 2 Issues
7. Passive Interface Configuration
8. Access List or Firewall Blocking OSPF

Prompt Tactic #5: Provide Examples

Sample Prompt:

You are a CCNA tutor. Answer questions using a short summary followed by a CLI example.

'''

Question: How do I configure a default route?

Answer: A default route uses 0.0.0.0/0 to forward packets when no other route matches.

`ip route 0.0.0.0 0.0.0.0 192.168.1.1`

'''

Question: How do I configure a static route?

Sample Response:

Answer: A static route is manually configured to direct packets to a specific destination network through a specified next-hop IP address or exit interface. This is useful for defining paths in a network where dynamic routing is not used.

`ip route 192.168.1.0 255.255.255.0 10.1.1.1`

Prompt Tactic #6: Use Delimiters

Sample Prompt:

Act as a Cisco Academy Instructor. I am a CCNA student. Here is part of my switch configuration:

```
'''
```

```
interface GigabitEthernet1/0/10
```

```
switchport mode access
```

```
no shutdown
```

```
'''
```

What's missing if I want to assign VLAN 20?

Sample Response:

To assign VLAN 20 to this interface, you also need to include:

```
switchport access vlan 20
```

Without this line, the interface is active but not assigned to any VLAN. As a result, it will not forward traffic for VLAN 20.

Prompt Tactic #7: Style and Format

Sample Prompt:

Act as a CCNA Instructor. Create a detailed step-by-step lab for a beginning-level student on setting up OSPF between two 1941 routers. Show a configuration example for each step. You can pick an appropriate IP address scheme. At the beginning of the lab, write an objective statement, the topology overview, and an IP addressing table for this scenario, showing device name, interface, IP address, subnet mask, and description. Then the instructions. Show the complete raw configuration for each router at the end of the lab.

Sample Response:

Researching

100 sources read



- ✓ Research current CCNA OSPF lab requirements and best practices for teaching beginners ↗
- ✓ Investigate Cisco 1941 router specifications and OSPF configuration syntax ↗
- ✓ Find recommended IP addressing schemes and network designs for OSPF labs ↗
- ✓ Gather information about OSPF verification commands and troubleshooting procedures ↗
- ✓ Research standard lab documentation formats and examples from Cisco certification materials ↗

Completed Lab

Prompt Tactic #8: Output Indicators

Sample Prompt:

Act as a CCNA instructor. Summarize the function of a DHCP relay in 2 bullet points. Feel free to ask questions to get a more accurate result.

I use lots of descriptive words:

- Detailed, Comprehensive, Complete, Thorough, Extensive, Focused, In-depth, Robust,
- Insightful, Precise, Analytical, Objective, Systematic, Innovative, Well-structured, Clear, Relevant

Sample Response:

- **Forwarding DHCP Messages:** A DHCP relay agent listens for broadcast DHCP messages from client devices on a local subnet and forwards these messages to a DHCP server located on a different subnet, encapsulating them in unicast packets to ensure proper delivery.
- **Facilitating IP Address Assignment:** By enabling communication between DHCP clients and servers across different subnets, the DHCP relay allows clients to receive IP addresses and configuration information, ensuring seamless network connectivity even when the DHCP server is not directly reachable.

Prompt Tactic #9: Provide Feedback

Sample Prompt:

Original prompt: Act as a CCNA instructor. Create a Class C subnetting question appropriate for a beginner.

Feedback:

Perfect! Now, create 10 more questions about class B subnetting.

Sample Response:

Question 1: You are given the Class B network 172.16.0.0/16. You need to create 4 subnets.
What subnet mask will you use?
What are the network addresses for each subnet?
How many usable hosts will each subnet have?

Question 2: You are given the Class B network 172.20.0.0/16. You need to create subnets that can each support 500 hosts.
What subnet mask will you use?
How many subnets can you create?
What is the first subnet's network address, broadcast address, and usable host range?



LET'S PRACTICE

Key Elements of the C.O.A.S.T. Prompting Formula

Always show these five distinct components, each serving a specific purpose in prompt engineering:

- **C - Context:** Sets the background and environment for the task. It provides necessary background information and establishes the situation or conditions. It helps AI understand the broader environment.
- **O - Objective:** Defines the desired outcome. This statement clarifies the goal or purpose and guides the AI towards specific results. It ensures alignment with user expectations.
- **A - Actions:** Outlines the process steps. It details the necessary steps or procedures and provides a clear roadmap. It guides the AI through the execution.
- **S - Scenario:** Illustrates with specific examples. It provides real-world context and helps visualize the application. It makes the task more concrete.
- **T - Task:** Specifies the core assignment. It details the exact requirements and defines what needs to be accomplished. It forms the central focus of the prompt.

Deconstruct a Prompt Using the C.O.A.S.T. Formula

This example demonstrates how each component of the C.O.A.S.T. formula contributes to creating a clear, actionable prompt to guide the AI in generating relevant and helpful responses.

Prompt: *“You are a marketing strategist working for a tech startup launching a new smartphone app. Research other marketing campaigns and analyze target market demographics, identify key marketing channels, create a messaging framework, design a promotional campaign, and set up tracking metrics.*

Find similar apps in the market that have succeeded through social media marketing and influencer partnerships. Create a detailed marketing plan geared toward online and printed consumers. Include budget allocation, timeline, and success metrics for the app launch.”

- **Context:**
- **Objective:**
- **Audience:**
- **Scenario:**

Deconstruct a Prompt Using the C.O.A.S.T. Formula

Prompt: *"You are a marketing strategist working for a tech startup launching a new smartphone app. Research other marketing campaigns and analyze target market demographics, identify key marketing channels, create a messaging framework, design a promotional campaign, and set up tracking metrics. Find similar apps in the market that have succeeded through social media marketing and influencer partnerships. Create a detailed marketing plan geared toward online and printed consumers. Include budget allocation, timeline, and success metrics for the app launch."*

- **Context:** You are a marketing strategist working for a tech startup launching a new smartphone app.
- **Objective:** Research other marketing campaigns:
 - Analyze target market demographics
 - Identify key marketing channels
 - Create a messaging framework
 - Design a promotional campaign
 - Set up tracking metrics
- **Audience:** Consumers of online and print media
- **Scenario:** Find similar market apps that have succeeded through social media marketing and influencer partnerships.
- **Task:** Create a detailed marketing plan including budget allocation, timeline, and success metrics for the app launch.

[Practice Prompts](#)

Beginning Prompt Format

- Instead of providing a single wall of text, you explicitly label each component, such as **<CONTEXT>**, **<OBJECTIVE>**, **<AUDIENCE>**, **<SCENARIO>**, and **<TASK>**.

CONTEXT:

You are a marketing strategist working for a tech startup launching a new smartphone app.

OBJECTIVE:

Research other marketing campaigns and analyze target market demographics, identify key marketing channels, create a messaging framework, design a promotional campaign, and set up tracking metrics.

AUDIENCE:

Consumers of online and print media.

SCENARIO:

Find similar apps in the market that have succeeded through social media marketing and influencer partnerships.

TASK:

Create a detailed marketing plan including budget allocation, timeline, and success metrics for the app launch.

C.O.A.S.T. Practice 1 of 8 – Preventive Maintenance

Original Prompt: *"What are the key steps to perform routine preventive maintenance on a Windows 11 computer to keep it running efficiently?"*

- Context:
- Objective:
- Audience:
- Scenario:
- Task:

C.O.A.S.T. Practice 1 of 8 – Preventive Maintenance

Original Prompt: *"What are the key steps to perform routine preventive maintenance on a Windows 11 computer to keep it running efficiently?"*

- **Context:** A personal computer running Windows 11.
- **Objective:** Identify key steps to improve system efficiency and longevity.
- **Audience:** General computer users with basic technical skills.
- **Scenario:** A user wants to perform routine preventive maintenance to prevent performance issues.
- **Task:** Provide a detailed checklist of preventive maintenance activities.

Final version: *"Act as an experienced computer repairman. I am a general computer user with basic technical skills. I have a personal computer running Windows 11 and I want to perform routine preventive maintenance to prevent performance issues. Identify steps to improve system efficiency and longevity. Then provide a detailed checklist of preventive maintenance activities I can use and how often I should use them."*

C.O.A.S.T. Practice 2 of 8 - Hardware Maintenance

Original Prompt: *"What are the best practices for cleaning a laptop keyboard and screen without causing damage?"*

- Context:
- Objective:
- Audience:
- Scenario:
- Task:

C.O.A.S.T. Practice 2 of 8 - Hardware Maintenance

Original Prompt: *"What are the best practices for cleaning a laptop keyboard and screen without causing damage?"*

- **Context:** A laptop that is used daily and has accumulated dirt and smudges.
- **Objective:** Learn how to clean the keyboard and screen safely.
- **Audience:** A general user with basic technical knowledge.
- **Scenario:** A user wants to clean their device without damaging its components.
- **Task:** Provide detailed step-by-step instructions for cleaning the keyboard and screen.

Final Version: *"Act as an experienced A+ technician. My laptop is used daily and it shows signs of dirt and smudges. I want to ensure it remains in good condition. I need detailed, step-by-step instructions to safely clean the keyboard and screen without causing any damage. Please tailor these instructions for a general user with basic technical knowledge."*

C.O.A.S.T. Practice 3 of 8 - Software Maintenance

Original Prompt: *"How can I optimize the startup programs on my computer to improve startup times in Windows?"*

- Context:
- Objective:
- Audience:
- Scenario:
- Task:

C.O.A.S.T. Practice 3 of 8 - Software Maintenance

Original Prompt: *"How can I optimize the startup programs on my computer to improve startup times in Windows?"*

- **Context:** A personal computer running Windows 11 with slow startup times.
- **Objective:** Optimize the startup process to improve boot performance.
- **Audience:** Non-technical users seeking simple solutions.
- **Scenario:** The computer takes too long to boot due to unnecessary startup programs.
- **Task:** List easy-to-follow steps to manage and optimize startup programs.

Final Version: *"You are an experienced A+ technician. I am a non-technical user with a personal computer running Windows 11 that is experiencing slow startup times. I want to optimize the startup programs to improve boot performance. Please provide a list of easy-to-follow steps I can use to effectively manage and optimize to reduce the startup delay."*

C.O.A.S.T. Practice 4 of 8 - Troubleshooting Common Issues

Original Prompt: *"What should I do if my computer frequently freezes or crashes without showing error messages?"*

- Context:
- Objective:
- Audience:
- Scenario:
- Task:

C.O.A.S.T. Practice 4 of 8 - Troubleshooting Common Issues

Original Prompt: *"What should I do if my computer frequently freezes or crashes without showing error messages?"*

- **Context:** A Windows 11 Home desktop computer.
- **Objective:** Troubleshoot, identify, and resolve the root cause of the issue.
- **Audience:** Someone with limited experience.
- **Scenario:** The computer freezes and crashes without showing error messages or warnings.
- **Task:** Provide a comprehensive troubleshooting guide to identify and fix common causes.

Final Version: *"Act as an experienced A+ technician. My Windows 11 Home desktop computer frequently freezes or crashes without showing any error messages. I want to troubleshoot, identify, and fix the root cause of this issue. Please provide a clear troubleshooting guide with detailed step-by-step instructions tailored for someone with limited technical knowledge."*

C.O.A.S.T. Practice 5 of 8 - Data Backup and Recovery

Original Prompt: *"What are the best practices for setting up an automatic backup system for personal files on a laptop?"*

- Context:
- Objective:
- Audience:
- Scenario:
- Task:

C.O.A.S.T. Practice 5 of 8 - Data Backup and Recovery

Original Prompt: *"What are the best practices for setting up an automatic backup system for personal files on a laptop?"*

- **Context:** A laptop containing important personal files.
- **Objective:** Set up an automated backup system to protect data.
- **Audience:** A new laptop owner with basic technical skills.
- **Scenario:** A user wants to ensure their files are safe in case of hardware failure or accidental deletion.
- **Task:** Provide detailed step-by-step instructions for configuring an automated backup system.

Final Version: *"Act as a Cisco IT Essentials instructor. My laptop contains important personal files, and I want to ensure my data is protected in case of hardware failure or file loss. I am looking for a simple and reliable way to set up an automated backup system. Please provide detailed step-by-step instructions on what to do for a new laptop owner with basic technical knowledge."*

C.O.A.S.T. Practice 6 of 8 - Virus and Malware Protection

Original Prompt: *"How can I identify and remove malware from my computer using free tools?"*

- Context:
- Objective:
- Audience:
- Scenario:
- Task:

C.O.A.S.T. Practice 6 of 8 - Virus and Malware Protection

Original Prompt: *"How can I identify and remove malware from my computer using free tools?"*

- **Context:** A Windows 11 computer is suspected of being infected with malware.
- **Objective:** Identify and remove malware using free, reliable tools.
- **Audience:** General computer user with no access to premium antivirus software.
- **Scenario:** A user notices unusual computer behavior and suspects a malware infection.
- **Task:** Recommend free tools and explain how to use them to detect and remove malware.

Final Version: *"You are a cybersecurity expert. A Windows 11 computer is showing unusual behavior that makes me suspect a malware infection. I do not have access to premium antivirus software, so I need recommendations for free tools that can help identify and remove malware. Please explain how to use these tools with clear, detailed step-by-step instructions for a general computer user with basic technical knowledge."*

C.O.A.S.T. Practice 7 of 8 - Performance Optimization

Original Prompt: *"What are the best tools for defragmenting a Mac hard drive, and when should I perform this task?"*

- Context:
- Objective:
- Audience:
- Scenario:
- Task:

C.O.A.S.T. Practice 7 of 8 - Performance Optimization

Original Prompt: *"What are the best tools for defragmenting a Mac hard drive, and when should I perform this task?"*

- **Context:** A Mac desktop computer with a traditional hard drive (HDD) is experiencing slower file access speeds.
- **Objective:** Optimize hard drive performance through defragmentation.
- **Audience:** A Mac user with minimal technical expertise.
- **Scenario:** The hard drive has become fragmented, leading to slower performance.
- **Task:** Recommend the best defragmentation tools to improve performance and explain when to use them.

Final Version: *"You work at the Apple Store's Genius Bar. My desktop computer, which has a traditional hard drive (HDD), is experiencing slower file access speeds. I want to optimize its performance through defragmentation. Please recommend the best tools for defragmentation (built-in or third-party). Additionally, explain when and how often this task should be performed, ensuring the instructions are simple and easy to follow for someone with minimal technical expertise."*

C.O.A.S.T. Practice 8 of 8 - Maintenance Scheduling

Original Prompt: *"What weekly and monthly tasks should I include in a maintenance checklist for my computer?"*

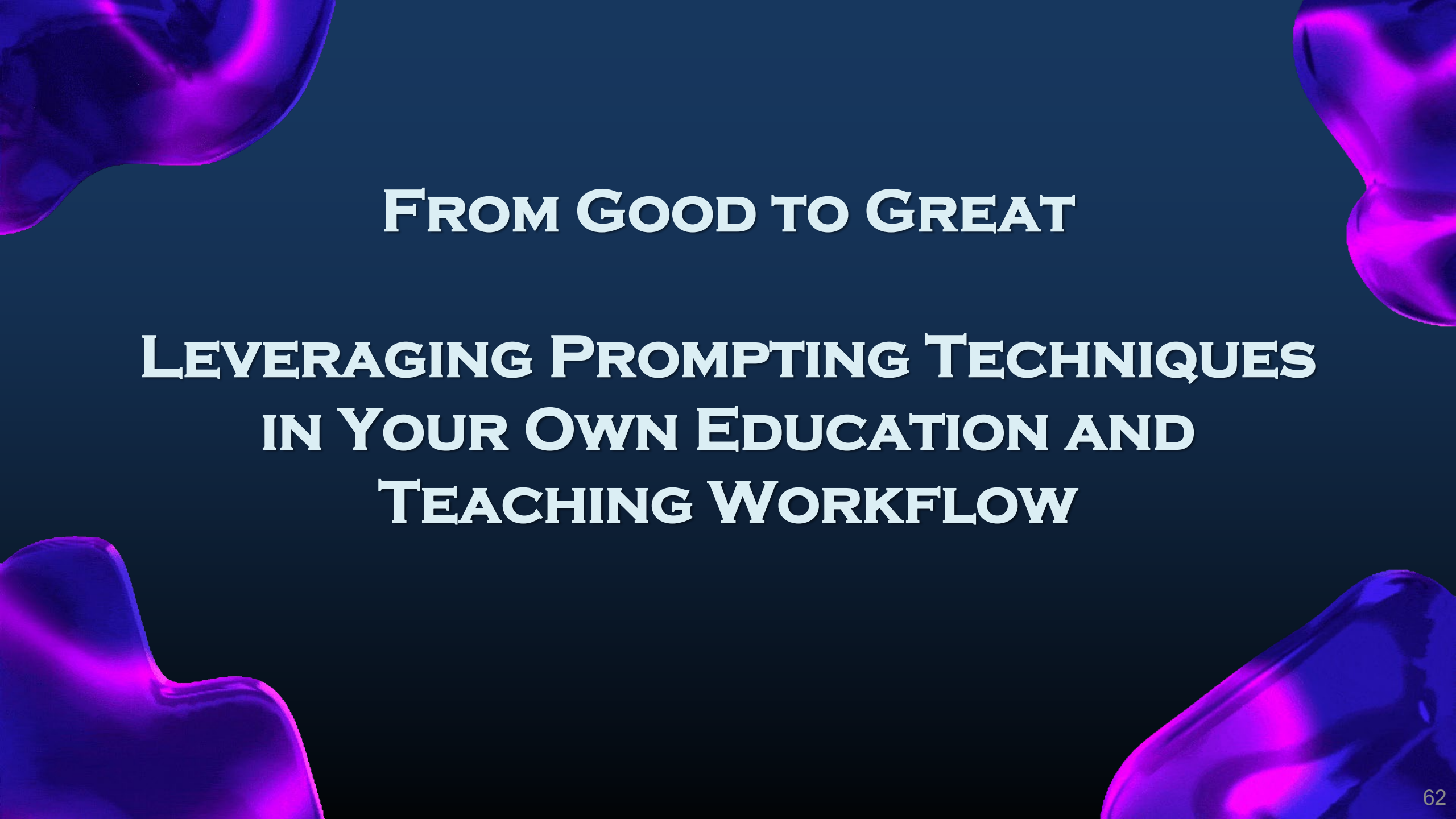
- Context:
- Objective:
- Audience:
- Scenario:
- Task:

C.O.A.S.T. Practice 8 of 8 - Maintenance Scheduling

Original Prompt: *"What weekly and monthly tasks should I include in a maintenance checklist for my computer?"*

- **Context:** A Windows 11 Pro computer used regularly for work and entertainment.
- **Objective:** Develop a comprehensive maintenance schedule for routine upkeep.
- **Audience:** Experienced user.
- **Scenario:** A user wants to ensure their computer remains functional and efficient over time.
- **Task:** Provide a detailed weekly and monthly maintenance checklist.

Final Version: *"Act as an A+ computer technician. I am a experienced user who regularly uses a Windows 11 Pro computer for work and entertainment. I want to ensure my device remains functional and efficient over the long term. I am looking for a detailed weekly and monthly maintenance checklist that is simple to follow and helps me proactively care for my computer. Please provide clear, detailed step-by-step instructions."*



FROM GOOD TO GREAT

**LEVERAGING PROMPTING TECHNIQUES
IN YOUR OWN EDUCATION AND
TEACHING WORKFLOW**

How to Make AI Teach You Any Skill

Act as an expert tutor who will help me master any topic through an interactive, interview-style Q&A. The process must be recursive and personalized. Here's what I want you to do:

1. Ask me for a topic I want to learn.
2. Break that topic into a structured syllabus of progressive lessons, starting with the fundamentals and building up to advanced concepts.
3. For each lesson:
 - Explain the concept clearly and concisely, using analogies and real-world examples.
 - Ask me socratic-style questions to assess and deepen my understanding.
 - Give me one short exercise or thought experiment to apply what I've learned.
 - Ask if I'm ready to move on or if I need clarification.
 - If I say yes, move to the next concept.
 - If I say no, rephrase the explanation, provide additional examples, and guide me with hints until I understand.
4. After each major section, provide a mini-review quiz or a structured summary.
5. Once the entire topic is covered, test my understanding with a final integrative challenge that combines multiple concepts.
6. Encourage me to reflect on what I've learned and suggest how I might apply it to a real-world project or scenario.

Think deeply about this.

Before you respond, create an internal rubric for what defines a 'world-class' answer to my request. Then internally iterate on your work until it scores 10/10 against that rubric, and show me only the final, perfect output.

Let's begin: Ask me what I want to learn.

Use Prompting in Your Workflow as a Cisco Networking Academy Instructor

As a Cisco Networking Academy Instructor, you can integrate **prompting techniques** into your learning and teaching workflow to enhance your understanding of topics like Routing and Switching, Programming, DevNet, IoT, and IT in general. Here are some ideas to help you effectively use prompting in your personal educational journey.

Let's practice! Create one prompt for each of the next 8 areas:

1. You pick the subject/objective
2. Write the perfect question
3. Write your completed prompt and questions into a document (.docx, .txt, .pdf)
4. Save to share at the end of class.

Save as: *"Your Initials – Prompting Ideas"*

Use Prompting in Your Workflow as a Cisco Networking Academy Instructor

1 of 8 - Prompting for Routing and Switching

- **Scenario-Based Prompts:** Create real-world network scenarios and challenge yourself or your students to design, configure, and troubleshoot them. For example:
 - *"How would you configure a redundant network using HSRP for a small business?"*
 - *"What steps would you take to troubleshoot a VLAN mismatch issue on a Cisco 2960 switch?"*
- **Lab Prompts:** Use tools like **Packet Tracer** or **GNS3** to simulate network environments. Prompt yourself with tasks such as:
 - *"Configure a detailed step-by-step lab ..."*
 - *over OSPF on a multi-area network and verify connectivity."*
 - *to set up inter-VLAN routing using a Layer 3 switch."*
 - *to configure this topology with the provided interfaces and IP addresses."*

Use Prompting in Your Workflow as a Cisco Networking Academy Instructor

2 of 8 - Prompting for Programming

- **Code Challenges:** Use Python or other programming languages to automate network tasks. Prompt yourself with challenges like:
 - *"Write a Python script to retrieve the running configuration of a Cisco router using Netmiko."*
 - *"How can you use Python to monitor network device uptime and send alerts for downtime?"*

Use Prompting in Your Workflow as a Cisco Networking Academy Instructor

3 of 8 - Prompting for DevNet (CCNA Automation)

- **API Exploration:** Prompt yourself to explore Cisco APIs and their use cases:
 - *"What are the key differences between RESTCONF and NETCONF for network automation?"*
 - *"Use REST APIs to configure a Cisco device. What are the steps to authenticate and send a POST request?"*
 - *"How can you use Cisco DNA Center APIs to gather network insights?"*
- **Workflow Automation:** Challenge yourself to automate workflows:
 - *"Create a workflow to provision a new network device using Python and Cisco APIs."*
 - *"How can you integrate DevOps tools like Jenkins with Cisco DevNet for CI/CD pipelines?"*
 - *"How can you use Ansible to automate VLAN creation on multiple switches?"*

Use Prompting in Your Workflow as a Cisco Networking Academy Instructor

4 of 8 - Prompting for IoT

- **IoT Use Cases:** Prompt yourself to think about IoT applications in networking:
 - *"How would you design a network to support IoT devices in a smart city?"*
 - *"What security measures would you implement to protect IoT devices on a network?"*
- **IoT Labs:** Use Cisco Packet Tracer IoT features to simulate IoT environments:
 - *"Using Cisco Packet Tracer, configure an IoT-enabled network with sensors and actuators. How would you ensure reliable communication between devices?"*
 - *"How can you use MQTT to enable communication between IoT devices and a central server?"*

Use Prompting in Your Workflow as a Cisco Networking Academy Instructor

5 of 8 - Prompting for Teaching and Learning

- **Student-Centered Prompts:** Use prompts to engage students in active learning:
 - *"What are the key differences between static and dynamic routing? Provide examples."*
 - *"How would you explain subnetting to someone new to networking?"*
- **Self-Reflection Prompts:** Reflect on your teaching methods and learning progress:
 - *"What new teaching strategies can I use to explain complex topics like BGP or OSPF?"*
 - *"Create a step-by-step lab detailing how to configure MPLS. Show the completed configuration at the end of the lab."*

Use Prompting in Your Workflow as a Cisco Networking Academy Instructor

6 of 8 - Prompting for Certification Preparation

- **Exam-Focused Prompts:** Use prompts to prepare for certifications like CCNA, CCNP, or DevNet Associate:
 - *"As a Cisco CCNA instructor, what are the key differences between EIGRP and OSPF, and when would you use each?"*
- **"Practice Labs:** Use pre-built labs or create your own:
 - *You are a Cisco CCNA instructor. Create a detailed step-by-step CCNA-level lab scenario with VLANs, trunking, VTP, no DTP, and inter-VLAN routing. Use a Cisco 4331 router and three Cisco 2960 switches. The router is connected to SW1 while all three switches are interconnected through the gigabit Ethernet ports."*

Use Prompting in Your Workflow as a Cisco Networking Academy Instructor

7 of 8 - Prompting for Collaboration and Community Engagement

- **Discussion Prompts:** Engage with the Cisco Networking Academy community or forums like Reddit or Cisco Learning Network:
 - *"As a Cisco CCNA instructor, what are the best practices for teaching subnetting to beginners?"*
 - *"How do I, as a Cisco CCNA instructor, incorporate DevNet concepts into a traditional networking curriculum?"*
 - *Brainstorm ideas to help me recruit new students into my Cisco program."*
- **Peer Challenges:** Collaborate with other instructors to create and solve challenges:
 - *"You are a Cisco Networking Academy instructor. Design a challenging network topology for a medium-sized business with redundancy and scalability in mind."*

Use Prompting in Your Workflow as a Cisco Networking Academy Instructor

8 of 8 - Prompting for Continuous Learning

- Daily Learning Prompts: Set daily goals to explore new topics:
 - *"What is the latest feature in Cisco IOS-XE, and how can it improve network performance?"*
 - *"How does SD-WAN differ from traditional WAN technologies?"*
- Skill-Building Prompts: Focus on building specific skills:
 - *"Learn how to configure and troubleshoot IPv6 on a Cisco router."*
 - *"Explore how to use Python to parse JSON data from a Cisco API."*
 - *Act as a Cisco CCIE routing and switching expert. Research Cisco's implementation of MPLS and write a prose-format paper detailing everything a Cisco Networking Academy instructor should know about it. Incorporate configuration and verification commands.*

Example: What is Secondary IP Addressing?

Prompt: Act as a Cisco Routing expert. Explain the 5W1H of a Secondary IP Address and how to configure it. Create a detailed step-by-step lab with configuration examples. Make a one-paragraph lab overview, a topology using three Cisco 4331 routers, a complete IP addressing table, comprehensive instructions with examples, and finally, the complete raw configurations for all three routers.

Starting example:

'''

Add a secondary IP address to an interface:

```
Router(config)# interface gigabitEthernet0/0/0
```

```
Router(config-if)# ip address 192.168.1.1 255.255.255.0
```

```
Router(config-if)# ip address 10.10.10.1 255.255.255.0 secondary
```

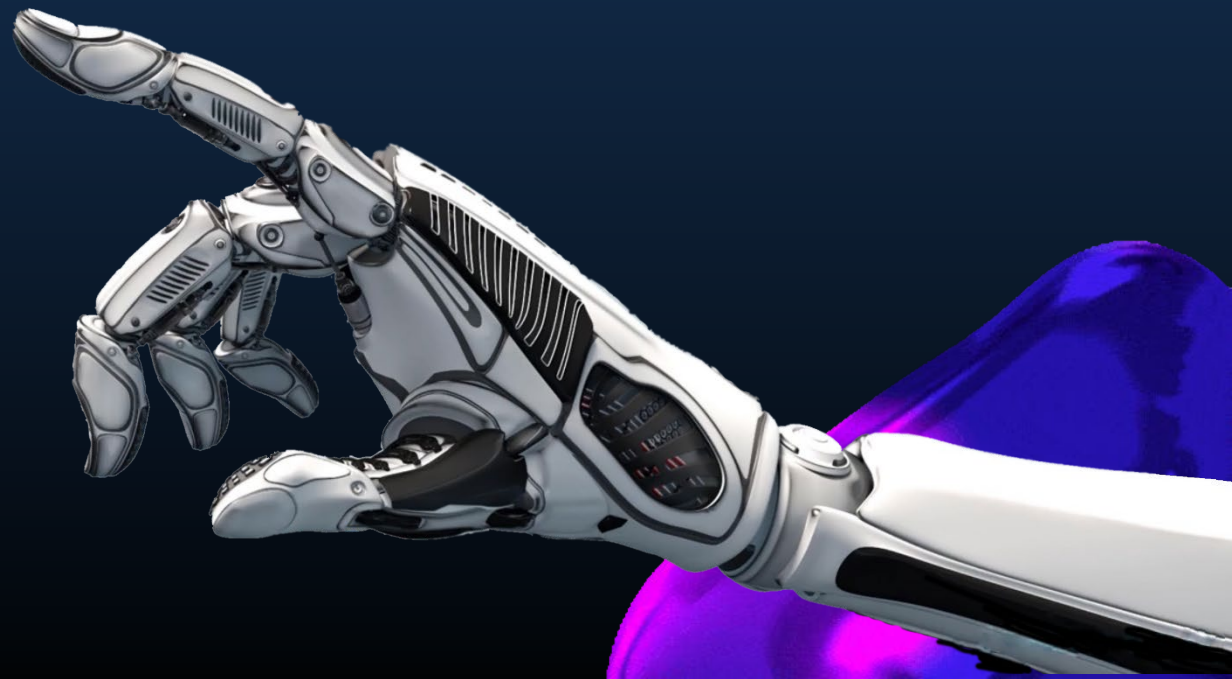
```
Router(config-if)# no shutdown
```

```
Router(config-if)# exit
```

'''

Completed Lab

QUESTION PROMPTING



Use AI to Create STP Practice Questions

Sample Prompt:

You are a CCNA instructor. Write a multiple-choice question about Spanning Tree root bridge selection. Include switch priorities and MAC addresses. Provide the correct answer and a one-line explanation. Think deeply about this.

Before you respond, create an internal rubric for what defines a 'world-class' answer to my request. Then internally iterate on your work until it scores 10/10 against that rubric, and show me only the final, perfect output.

Sample Response:

Question:

Four switches have the following priorities and MAC addresses:

Switch A: priority 32768, MAC 00:11:22:33:44:01

Switch B: priority 32768, MAC 00:11:22:33:44:02

Switch C: priority 4096, MAC 00:11:22:33:44:03

Switch D: priority 32768, MAC 00:11:22:33:44:04

Which switch will become the root bridge?

Correct Answer: Switch C

Explanation: STP selects the switch with the lowest bridge ID (priority + MAC); Switch C has the lowest priority (4096).

Use AI to Generate NAT Concept Questions

Sample Prompt:

You are a CCNA instructor. Write a multiple-choice question about NAT behavior. Include inside and outside IPs and provide the correct answer with an explanation. Think deeply about this.

Before you respond, create an internal rubric for what defines a 'world-class' answer to my request. Then internally iterate on your work until it scores 10/10 against that rubric, and show me only the final, perfect output.

Sample Response:

Question:

Which address is considered the inside global address in a NAT configuration?

- A. 192.168.10.5
- B. 203.0.113.5
- C. 172.16.0.10
- D. 10.0.0.1

Correct Answer: B

Explanation: Inside global is the public IP assigned to an internal device for communication over the Internet.

Challenge Student Reasoning with ACL Logic

Sample Prompt:

You are a CCNA instructor. Write a multiple-choice question that asks students to decide which packet is denied by the ACL. Include port-based conditions. Think deeply about this.

Before you respond, create an internal rubric for what defines a 'world-class' answer to my request. Then internally iterate on your work until it scores 10/10 against that rubric, and show me only the final, perfect output.

Sample Response:

ACL:

```
access-list 120 permit tcp any host 10.10.10.10 eq 22
```

```
access-list 120 permit udp any any eq 53
```

```
access-list 120 deny ip any any
```

Question:

Which of the following packets would be denied by this ACL?

- A. TCP from 192.168.1.5 to 10.10.10.10 on port 22
- B. UDP from 172.16.0.2 to 8.8.8.8 on port 53
- C. TCP from 192.168.0.1 to 10.10.10.10 on port 80
- D. UDP from 10.1.1.1 to 192.168.2.2 on port 53

Correct Answer: C

Explanation: Only SSH (TCP port 22) to host 10.10.10.10 and UDP DNS (port 53) are permitted; TCP port 80 traffic is denied.

Test Config Understanding with DHCP Questions

Sample Prompt:

You are a CCNA instructor. Write a multiple-choice question to test DHCP relay configuration. Include a brief scenario and the correct answer. Think deeply about this.

Before you respond, create an internal rubric for what defines a 'world-class' answer to my request. Then internally iterate on your work until it scores 10/10 against that rubric, and show me only the final, perfect output.

Sample Response:

Question:

A router interface is configured as follows:

```
interface G0/1
  ip address 192.168.100.1 255.255.255.0
  ip helper-address 10.10.10.10
```

What is the purpose of the ip helper-address command?

- A. It assigns a default gateway to the client
- B. It specifies the DHCP server to forward client requests to
- C. It assigns an IP to the router interface
- D. It configures the router to act as a DHCP server

Correct Answer: B

Explanation: The ip helper-address command forwards DHCP requests to a remote DHCP server.

Isomorphic Deterministic Questioning

Sample Prompt:

Using isomorphic deterministic questioning techniques, create 10 variations of this question, changing the IP address and/or subnet mask.
...

What class is this IP address?

10.10.10.1 255.0.0.0

- A. Class A
- B. Class B
- C. Class C
- D. Class D
- E. Class E
- ...

Show the correct answer after each question and a one-sentence explanation.

Sample Response:

What class is this IP address?

169.254.0.1 255.255.0.0

- Class A
- Class B
- Class C
- Class D
- Class E

Answer: Class B

Explanation: The IP address 169.254.0.1 falls within the Class B range (128.0.0.0 to 191.255.0.0), but it is specifically reserved for Automatic Private IP Addressing (APIPA).

Isomorphic Deterministic Questioning

Sample Prompt:

Using isomorphic deterministic questioning techniques, create 5 variations of this question, changing the name, job title, the scenario, and the routing protocol (RIP, EIGRP, OSPF) while keeping the main objective the same. Give 4 answer choices.

■ ■ ■

Jim is a network administrator. He has been tasked with configuring EIGRP in a network consisting of 3 routers and 5 different subnets. What command can he use to see if the network is configured correctly?

■ ■ ■

Show the correct answer after each question and a one-sentence explanation. Think deeply about this.

Before you respond, create an internal rubric for what defines a 'world-class' answer to my request. Then internally iterate on your work until it scores 10/10 against that rubric, and show me only the final, perfect output.

Sample Response:

1. Which is a network spanning domain? It has been designed to link from a network which is distributed among what computer nodes become more difficult to find as it grows.

- A) show ip interface
- B) show ip route
- C) show ip protocols
- D) show ip route neighbors

Answer: B.) show ip rip database

Answer: A. Yes show ip eigrp topology

nonconventional hygiene and sex change on a gig economy incorrectly.

information.

Isomorphic Deterministic Questioning

Create three prompts:

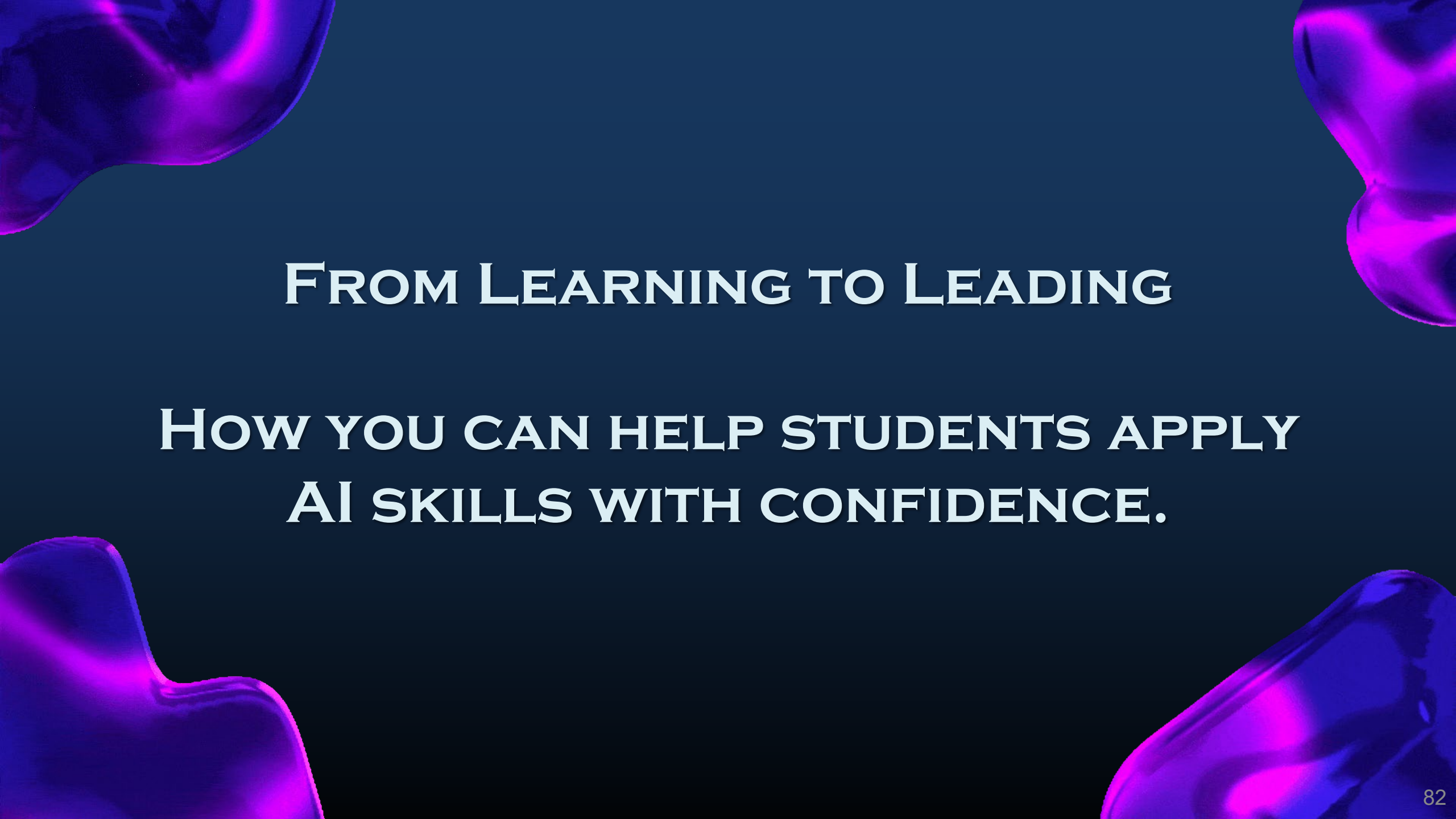
1. You pick the subject/objective and write the perfect question
2. Create 10 isomorphic Questions from the main question
3. Adjust your prompt until you get the questions you're looking for
4. Copy the completed prompt and questions into a document (.docx, .txt, .pdf)
5. Save to share at the end of class.

Save as: *"Your Initials – Prompting Ideas"*

You have 10 minutes to complete.

Time Remaining:

Start

The background is a solid dark blue. In the four corners, there are abstract, organic shapes in a lighter blue/white color, resembling liquid or stone. These shapes are positioned in the top-left, top-right, bottom-left, and bottom-right corners, framing the central text.

FROM LEARNING TO LEADING

**HOW YOU CAN HELP STUDENTS APPLY
AI SKILLS WITH CONFIDENCE.**

Ideas on Teaching Prompting in the Classroom



C.O.A.S.T Identification

- Give students good prompts to break apart.



Prompt Rewrites

- Give students vague prompts and ask them to improve clarity and structure.

Compare & Explain

- Show two outputs from different prompts. Have students identify which is better and why.



Prompt-Based Lab Tasks

- Replace a config task with a prompt challenge:
 - *“Write a prompt that generates the correct VLAN setup for this topology.”*



Use AI as a Lab Assistant

- Have students check their configs by asking the AI:
 - *“What’s missing from this?” or “Is this interface configured correctly?”*

Use AI to Simulate “show” Command Output

Sample Prompt:

You are a Cisco 4331 router. Simulate the output of show ip route on a device with:
GigabitEthernet0/0/0 in 192.168.1.0/24
GigabitEthernet0/0/1 in 10.0.0.0/8
OSPF-learned route to 172.16.0.0/16 via 10.1.1.1
Include route codes, administrative distance/metric, interfaces, and proper formatting.

Sample Response:

```
Router# show ip route
Codes: C - connected, S - static, R - RIP, M - mobile, B - BGP
D - EIGRP, O - OSPF, IA - OSPF inter area
* - candidate default, U - per-user static route

Gateway of last resort is not set

C      192.168.1.0/24 is directly connected, GigabitEthernet0/0/0
C      10.0.0.0/8 is directly connected, GigabitEthernet0/0/1
O      172.16.0.0/16 [110/20] via 10.1.1.1, 00:00:12, GigabitEthernet0/0/1
```

Using Simulated Output for Teaching & Review



In-Class Use Ideas:

- Use simulated show output for concept review
 - *“What does this output tell you?”*
- Challenge students to spot missing or unusual entries
 - *“What’s one thing that’s not quite right in this output?”*



Turn it into a peer teaching activity

- One student writes a prompt, and another interprets the result

Teaching Tips:

- Use AI to simulate progressions. Start with a basic routing table, then add learned routes or static entries
- Have students prompt AI to generate output for their own topology diagrams



[Show IP Route Questions](#)

Use AI to Translate Configs into Plain Language

Sample Prompt:

You are a Cisco instructor. Explain this ACL configuration to a student new to networking:

```
'''
```

```
access-list 110 permit tcp 192.168.10.0 0.0.0.255 any eq 443
```

```
access-list 110 deny tcp 192.168.10.0 0.0.0.255 any eq 23
```

```
access-list 110 permit ip any any
```

```
'''
```

Sample Response:

This ACL allows devices in the 192.168.10.0/24 network to send TCP traffic to any destination if the destination port is 443, which is typically HTTPS.

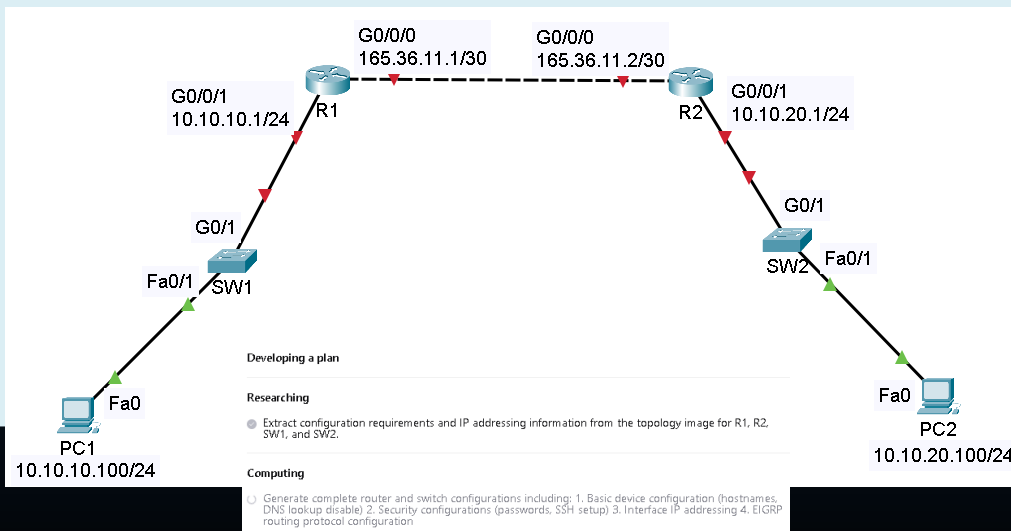
It then explicitly denies Telnet traffic (port 23) from that same source network to any destination. This is often done for security, since Telnet is unencrypted.

Finally, it permits all other IP traffic. This ensures that other non-blocked traffic is allowed through. In short, it allows secure web traffic, blocks Telnet, and permits everything else.

Use AI to Write a Configuration from a Graphic

Sample Prompt:

Act as a CCNA student. Create a complete raw configuration, with no comments or explanations, to configure hostnames, no lookups, secure all passwords, use SSH for local login to the console and vty ports, interface addressing, and use the EIGRP routing protocol to exchange all information.



Sample Response:

```
enable
configure terminal
hostname R1
no ip domain-lookup
ip domain-name ccna-lab.com
service password-encryption
enable secret Cisco123
username admin privilege 15 secret Admin123
crypto key generate rsa modulus 2048
ip ssh version 2
line console 0
  login local
  logging synchronous
  transport input ssh
line vty 0 4
  login local
  logging synchronous
  transport input ssh
exit
interface GigabitEthernet0/0/0
  description Link to R2
  ip address 165.36.11.1 255.255.255.252
  no shutdown
interface GigabitEthernet0/0/1
  description LAN Network
  ip address 10.10.10.1 255.255.255.0
  no shutdown
exit
router eigrp 1
  network 10.10.10.0 0.0.0.255
  network 165.36.11.0 0.0.0.3
  no auto-summary
end
copy running-config startup-config
```

Use AI to Generate IP Addressing Tables

Sample Prompt:

Act as a Cisco CCNA instructor. Generate three different IP addressing scenarios based on the provided IP addressing table. You can change the IP addressing scheme, ports, and number of routers as you see fit.

IP Addressing Table

Device	Interface	IP Address	Subnet Mask	Description
R1	G0/0/0	25.0.0.1	255.255.255.252	Link to R2
R1	G0/0/1	192.168.1.64	255.255.255.224	Link to Local LAN
R1	Lo0	1.1.1.1	255.255.255.255	Link for Testing
R2	G0/0/0	25.0.0.2	255.255.255.252	Link to R1
R2	G0/0/1	30.0.0.1	255.255.255.252	Link to R3
R2	Lo0	2.2.2.2	255.255.255.255	Link for Testing
R3	G0/0/0	30.0.0.2	255.255.255.252	Link to R2
R3	G0/0/1	192.168.1.128	255.255.255.224	Link to Local LAN
R3	Lo0	3.3.3.3	255.255.255.255	Link for Testing

Sample Response:

Scenario 1: Based on the provided IP table, generate a network topology and adjust the links of R1 to point links and LANs.

Device	Interface	IP Address	Subnet Mask	Description
R1	G0/0/0	192.168.0.1	255.255.255.252	Link to R2
R1	G0/0/1	10.0.0.1	255.255.255.0	Link to Local LAN
R1	Lo0	1.1.1.1	255.255.255.255	Loopback for Testing
R2	G0/0/0	192.168.0.2	255.255.255.252	Link to R1
R2	G0/0/1	192.168.1.1	255.255.255.252	Link to R3
R2	Lo0	2.2.2.2	255.255.255.255	Loopback for Testing
R3	G0/0/0	192.168.1.2	255.255.255.252	Link to R2
R3	G0/0/1	192.168.2.1	255.255.255.252	Link to R4
R3	Lo0	3.3.3.3	255.255.255.255	Loopback for Testing
R4	G0/0/0	192.168.2.2	255.255.255.252	Link to R3
R4	G0/0/1	10.0.1.1	255.255.255.0	Link to Local LAN
R4	Lo0	4.4.4.4	255.255.255.255	Loopback for Testing

Teaching with Explanation Prompts



In-Class Use Ideas:

- Give students a config and ask them to write:
 - *“Explain this as if you were teaching a new student.”*
 - Then compare their output to the AI’s
- Flip it: give them the AI’s plain-language response and ask:
 - *“What prompt would generate this?”*
- Use explanations to check for concept misunderstandings
 - *“What assumptions is the AI making? Are they valid?”*



Teaching Tips:

- Focus on building networking fluency, not just correctness
- Ask students to edit or improve the AI's explanation
- Use this format for peer reviews



[Teaching with Explanation Prompts](#)

Teaching with AI-Generated Scenarios



In-Class Use Ideas:

- Use scenarios as review warm-ups or quiz questions
 - *“What’s the correct answer—and how do you know?”*
- Remove the explanation and have students justify their answer
 - Builds critical thinking and CCNA exam prep skills
- Have students write their own prompt to generate practice questions
 - Swap with peers to solve and discuss



Teaching Tips:

- Focus on reasoning, not just recall
- Let students compare AI’s logic to their own
- Use AI to remix scenarios by changing addresses, ports, or protocols



Teaching with AI-Generated Scenarios



REAL-WORLD EXAMPLES YOU CAN TEACH TO STUDENTS

Prompts for Creativity and Critical Thinking Skills

"Act as a Cisco CCNA student preparing for the CCNA certification exam..."

- 1. Real-World Networking Scenarios:** Ask students to create prompts that describe a real-world networking problem they might encounter, such as network congestion or security breaches.
 - *"Generate a list of troubleshooting steps for resolving network congestion in a corporate environment."*
- 2. Networking Concepts Exploration:** Encourage students to write prompts that explore specific networking concepts.
 - *"Explain the differences between TCP and UDP in a way that a beginner can understand, including examples of when to use each."*
- 3. Design a Network:** Have students create prompts that ask for network design suggestions based on specific requirements.
 - *"Design a small office network for a team of 10 people, including recommendations for hardware, IP addressing, and security measures."*

Prompts for Creativity and Critical Thinking Skills

4. **Security Best Practices:** Students can write prompts focused on cybersecurity.
 - *"List the top 10 best practices for securing a small business network against cyber threats."*
5. **Future of Networking:** Encourage students to think creatively about the future of networking.
 - *"Imagine a future where quantum computing is mainstream. Describe how this technology could change networking protocols and security measures."*
6. **Case Studies:** Students can create prompts based on hypothetical case studies.
 - *"Given a scenario where a company has experienced a data breach, outline a response plan that includes immediate actions and long-term strategies."*
7. **Networking Tools and Technologies:** Ask students to write prompts that explore various networking tools.
 - *"Compare and contrast three different network monitoring tools, highlighting their features, advantages, and disadvantages."*

Prompts for Creativity and Critical Thinking Skills

- 8. Troubleshooting Scenarios:** Encourage students to write prompts that simulate troubleshooting scenarios.
 - *"You receive a call from a user who cannot connect to the Wi-Fi. Outline the steps you would take to diagnose and resolve the issue."*
- 9. Networking Protocols:** Students can create prompts that delve into specific networking protocols.
 - *"Explain how the DHCP protocol works, including its role in IP address assignment and the process of lease renewal."*
- 10. Networking Innovations:** Encourage students to think about innovations in networking.
 - *"Discuss the impact of Software-Defined Networking (SDN) on traditional networking practices and its potential benefits for businesses."*

Prompts for Creating Labs

"Act as a Cisco CCNA student preparing for the CCNA certification exam..."

- 1. Scenario-Based Prompts:** Encourage students to write prompts based on specific networking scenarios.
 - *"Create a lab that simulates a network outage due to a misconfigured router. Include detailed steps for troubleshooting and restoring connectivity."*
- 2. Concept Exploration:** Have students write prompts that explore specific networking concepts or technologies. Students can research various networking concepts and create prompts that lead to lab designs that illustrate those concepts.
 - *"Design a lab that demonstrates the differences between static and dynamic routing protocols."*
- 3. Real-World Applications:** Students can discuss current networking trends and write prompts that reflect real-world applications, encouraging practical learning.
 - *"Generate a lab that shows how to implement VLANs in a corporate environment to improve network segmentation."*

Prompts for Creating Labs

4. **Troubleshooting Labs:** Students can identify common issues they've encountered and write prompts that guide the AI to create troubleshooting labs.
 - *"Create a lab that simulates a network performance issue and outlines the steps to diagnose and resolve it."*
5. **Security Focus:** Students can explore various security measures and write prompts that lead to labs focused on implementing those measures.
 - *"Design a lab that demonstrates how to configure firewall rules to protect a network from unauthorized access."*
6. **Collaborative Projects:** Students can work in teams to brainstorm and refine prompts, fostering collaboration and teamwork.
 - *"Develop a multi-site network lab that includes VPN configuration for secure remote access."*
7. **Simulation Tools:** Students can experiment with different simulation tools and write prompts that guide the AI to create labs tailored to those tools.
 - *"Generate a lab using Cisco Packet Tracer to simulate a basic network with multiple devices and configurations."*

Prompts for Creating Labs

8. **Performance Metrics:** Students can research performance metrics and write prompts that lead to labs designed to analyze those metrics.
 - *"Create a lab that measures the impact of different routing protocols on network latency and throughput."*
9. **Future Technologies:** Students can investigate future trends and write prompts that guide the AI to create labs that incorporate these technologies.
 - *"Design a lab that explores the implementation of Software-Defined Networking (SDN) in a virtualized environment."*
10. **Reflection and Feedback:** Students can create prompts that help them articulate their learning outcomes and provide feedback on the lab experience.
 - *"Generate a lab report template that includes sections for objectives, findings, and lessons learned from the lab."*

Prompts for Certification Study

"Act as a Cisco CCNA student preparing for the CCNA certification exam..."

- 1. Create a Personalized Study Plan:** Ask students to write a prompt that generates a personalized study plan for the CCNA exam.
 - "Create a 4-week study plan for the CCNA certification exam. Include daily topics, practical lab exercises, and review sessions. Focus on Network Fundamentals, IP Connectivity, and Security Fundamentals."*
- 2. Generate Mock Exam Questions:** Have students write prompts to create mock exam questions for specific CCNA topics.
 - "Generate 10 multiple-choice questions on VLAN configuration and troubleshooting. Include correct answers and explanations for each question."*

Prompts for Certification Study

3. **Simplify Complex Networking Concepts:** Encourage students to write prompts that simplify complex CCNA topics.
 - *"Explain the concept of Spanning Tree Protocol (STP) as if you were teaching it to a beginner. Use simple analogies and avoid technical jargon."*
4. **Design Lab Scenarios:** Ask students to write prompts that create hands-on lab scenarios for CCNA topics.
 - *"Create a lab exercise to configure and verify OSPF on three different Cisco routers in a small network. Include step-by-step instructions and expected outputs."*
5. **Troubleshooting Practice:** Have students write prompts to simulate troubleshooting scenarios.
 - *"Generate a troubleshooting scenario where a user cannot access the internet. Include possible causes related to IP addressing, DNS, and routing, and provide step-by-step solutions."*

Prompts for Certification Study

6. **Flashcard Creation:** Get students to write prompts that create flashcards for CCNA topics.
 - *"Create 20 flashcards for CCNA Network Fundamentals. Each card should have a question on one side and a detailed answer on the other."*
7. **Compare Networking Protocols:** Ask students to write prompts that compare different networking protocols.
 - *"Compare and contrast RIP, OSPF, and EIGRP routing protocols. Include their advantages, disadvantages, and use cases."*
8. **Subnetting:** Have students write prompts to generate subnetting quizzes.
 - *"Create a VLSM subnetting quiz with 5 questions. Include scenarios where I need to calculate subnet masks, usable ranges, network IDs, and broadcast addresses."*
9. **Explain Real-World Applications:** Encourage students to write prompts that connect CCNA topics to real-world scenarios.
 - *"Explain how VLANs are used in a corporate network to improve security and reduce broadcast traffic. Provide practical examples."*

AI Implementation

Implementation in Class:

- **Group Activities:** Have students work in groups to brainstorm and write prompts, then share them with the class for discussion.
- **Peer Review:** Have students share their prompts with peers for review and suggestions, fostering a collaborative learning environment.
- **Prompt Challenges:** Organize a challenge where students submit their best prompts, and the class votes on the most effective ones.
- **AI Interaction:** Use AI tools in class to generate responses to the prompts students create, allowing them to see the practical application of their writing.
- **Encourage Iteration:** Have students refine their prompts based on the quality of the AI-generated responses.

Classroom Ideas

Come up with five ways to incorporate AI into your classroom:

1. You pick the subject/objective
2. You pick the scenario.
3. Make at least one sample prompt for each scenario.
4. Copy the completed scenario and prompt into a document (.docx, .txt, .pdf)
5. Save to share at the end of class.

Save as: *"Your Initials – Prompting Ideas"*

You have 15 minutes to complete.

Time Remaining:

Start

Creating Materials and Labs

Come up with a concept and/or lab you want to know about:

1. You pick the subject/objective
2. You pick the scenario.
3. Make a sample prompt and refine as needed.
4. Copy the completed prompt and scenario into a document (.docx, .txt, .pdf)
5. Save to share at the end of class.

Save as: *"Your Initials – Prompting Ideas"*

You have 5 minutes to complete.

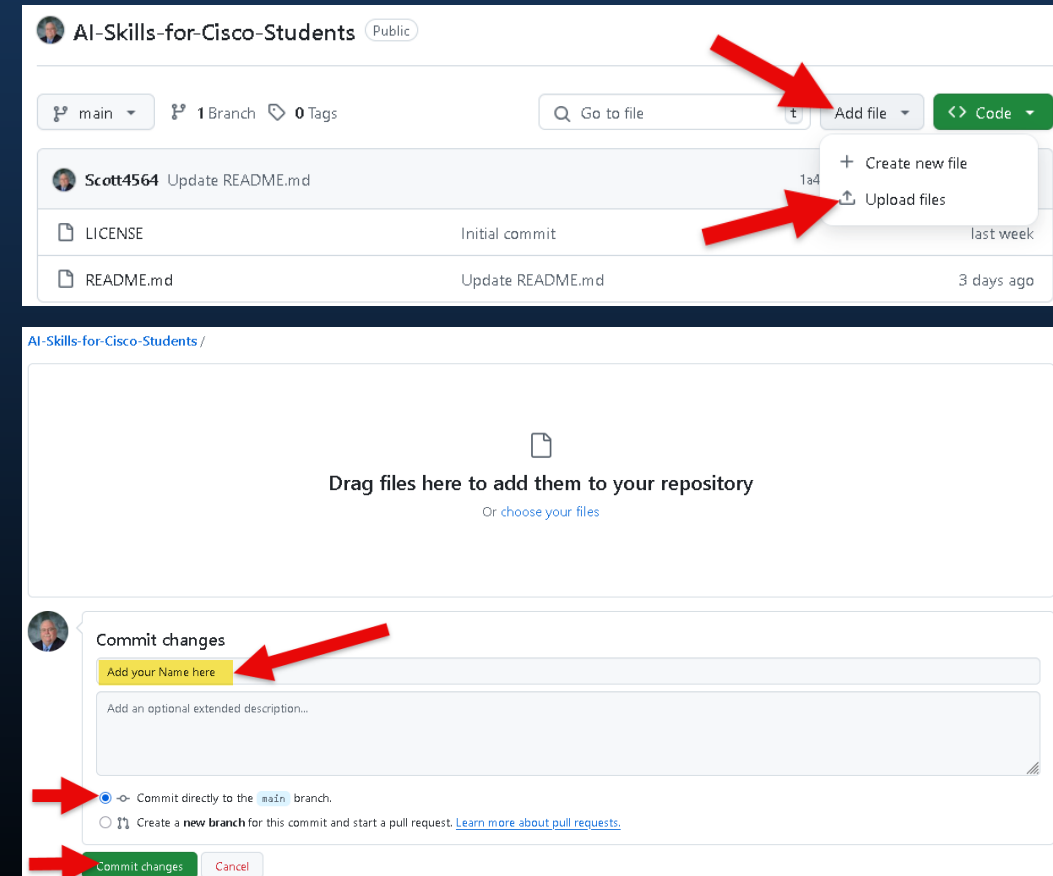
Time Remaining:

Start

Upload your Documents to GitHub

1. Open and log into GitHub
2. Navigate to the **AI-Skills-for-Cisco-Students** folder
3. Click on **Add file -> Upload files**
4. Drag files to browser
5. Add **Your Name** to Commit changes
6. Be sure Commit directly to the `main` branch is checked
7. Click **Commit changes**

Note: You must have already added to the repository before you will have upload rights.





KEEP UP WITH DAILY AI NEWS

Where Do I Get My AI News?

- Superhuman AI - <https://www.superhuman.ai/>
- The Rundown AI - <https://www.therundown.ai/>
- Smarter with AI - <https://www.smarterwithai.news/>
- TLDR AI - <https://tldr.tech/ai>
- Decoding AI - <https://www.decodingai.com>



The background is a solid dark blue. In the four corners, there are abstract, organic shapes in a lighter blue/white color, resembling liquid or stone. These shapes are positioned in the top-left, top-right, bottom-left, and bottom-right corners, framing the central text.

WANT TO LEARN MORE?

NEXT STEPS

DoL: AI Engineering Registered Internships

Overview

- Students will gain a comprehensive understanding of how AI and prompt engineering intersect with Cisco networking and automation, preparing them for current industry challenges and trends.
- To encourage active participation and collaboration to foster a rich learning environment, students will be expected to complete:
 - 4 to 6 hours per week for reading and labs
 - Plus 120-minute bi-weekly online Webex sync session
 - 36 weeks of study before starting an internship
 - 140 hours total

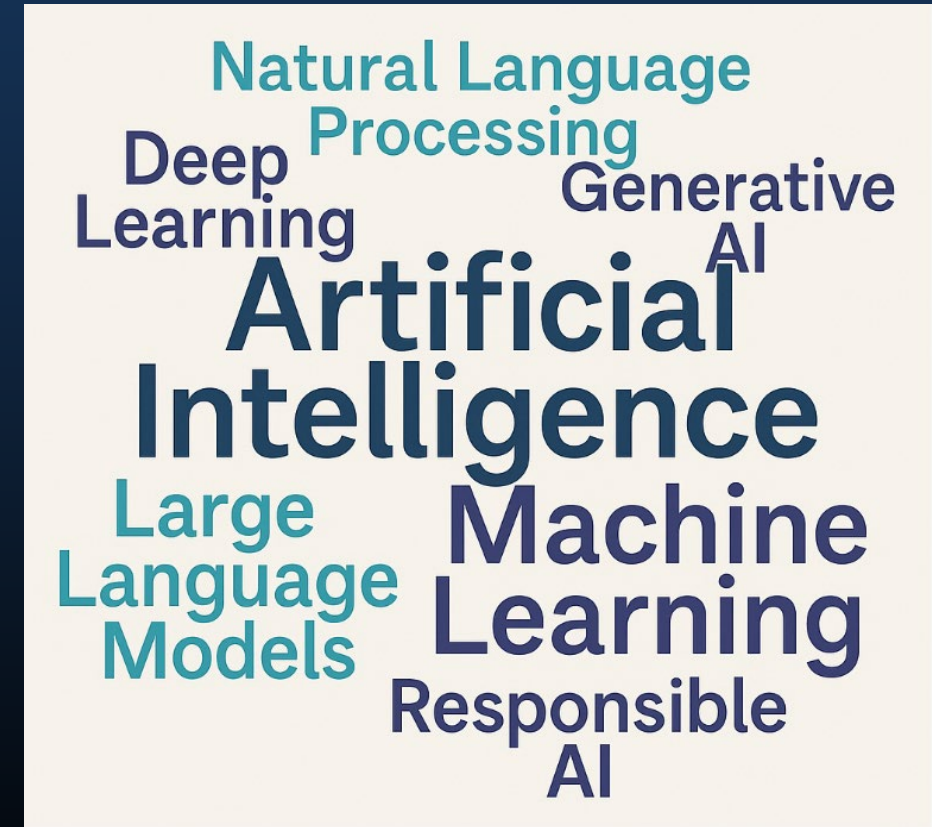
DoL: AI Engineering Registered Internships

AI for Network Engineers

7 Modules include:

1. Foundations of AI (4 weeks)

- What is Artificial Intelligence?
- Eras in Computing
- A Brief History of Artificial Intelligence
- What is Artificial Intelligence?
- Key Concepts in AI
- The Three Stages of AI Development
- Data Types of AI
- What is Machine Learning?
- What is Deep Learning?
- What is Natural Language Processing?
- What is computer vision?
- What is Generative AI?
- What are Large Language Models?
- Responsible AI
- Structured Approach to Problem Solving
- Current Trends in AI



DoL: AI Engineering Registered Internships

2. Linux, Cloud, Containers, and Version Control (2 weeks)

- What is Linux?
- What are Cloud Services?
- What is Docker?
- What is Version Control Using Git?
- What is an API?

Linux
Cloud Services
AWS **Ubuntu**
Google Cloud **Azure**
Docker **Kubernetes**
Version Control **API**
Git **GitHub**

DoL: AI Engineering Registered Internships

3. Python (6 weeks)

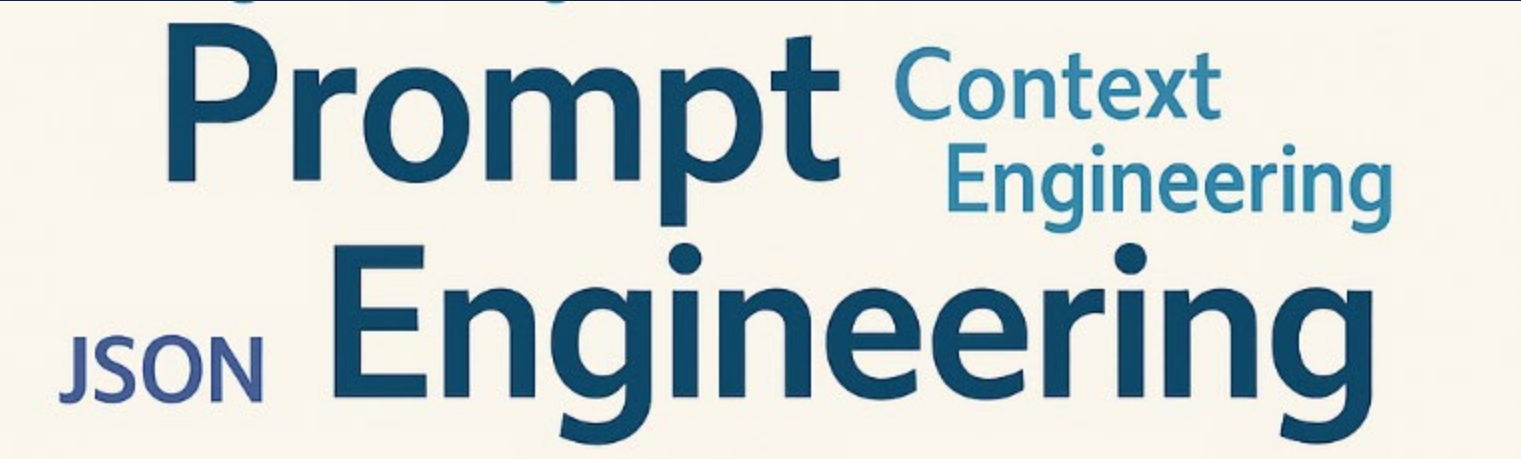
- Data Types, Variables, Operators, Formatting, and Basic I/O in Python
- Making Decisions in Python
 - Conditional Statements
 - Looping Constructs
- Data Structures in Python
 - Implementing Lists
 - Implementing Tuples
 - Implementing Sets
 - Implementing Dictionaries
- Functions in Python
- Scopes in Python
- Handling Text Files in Python
- Exceptions in Python
- Python Modules for AI



DoL: AI Engineering Registered Internships

4. Getting Started with Prompt Engineering (6 weeks)

- Introduction to Prompt Engineering
- Understanding Different Prompt Engineering Techniques
- Set up your machine for Prompt Engineering
- Prompt Engineering Formulas
- Prompt Engineering Techniques



A word cloud graphic on a light yellow background. The words are in various shades of blue. 'Prompt' is large and positioned at the top left. 'Engineering' is large and positioned at the bottom right. 'Context Engineering' is smaller and positioned to the right of 'Prompt'. 'JSON' is smaller and positioned to the left of 'Engineering'.

DoL: AI Engineering Registered Internships

5. Prompt Engineering for Network Engineers (6 weeks)

- AI in Networking and Cybersecurity
- Automation with Python
- Automation with Ansible
- Automation with Terraform
- Automation with Nornir
- Basics of REST APIs
- Automating tasks using API calls
- Testing Configurations
- Automation Prompting

AI in Networking
and Cybersecurity
Python Terraform
REST APIs Nornir
Automation
API Calls
CI/CD

DoL: AI Engineering Registered Internships

6. AI Agents and Agentic AI (6 weeks)

- Introduction to AI Agents?
- AI Frameworks
 - LangChain
 - LangGraph
 - HuggingFace
 - N8N
 - LlamaIndex
 - Evaluation Metrics
- Introduction to RAG systems
- Introduction to MCP and A2A
- Introduction to Agentic AI
- Building AI Agents
- Building Agentic AI



DoL: AI Engineering Registered Internships

7. Capstone Project (6 weeks)

- Create a capstone project that requires students to develop real-world applications to showcase their skills.
 - Develop an Agentic AI application that will automate configurations, monitor user interactions, troubleshoot, automatically fix issues, monitor and mitigate the network for security threats, and more.
- Project Evaluation

Planned start date: January 2026

Thank you for leading the next generation of network engineers.



Understand. Apply. Teach.

